

	BIORAD MEDISYS PVT. LTD.	Document No.	CER/DJS/61
		Revision. No	01
	CLINICAL EVALUATION REPORT PRODUCT- DOUBLE J STENT	Revision Date	26-12-2025

Double J Stent (Short Term)

Clinical Evaluation Report

	BIORAD MEDISYS PVT. LTD.	Document No.	CER/DJS/61
		Revision. No	01
	CLINICAL EVALUATION REPORT PRODUCT- DOUBLE J STENT	Revision Date	26-12-2025

Table of contents

1	Summary.....	8
2.1	Identification of device(s)	11
2.2	Name and address of the manufacturer	11
2.3	Administrative particulars.....	12
2.4	Device description.....	14
2.5	Device classification	14
2.6	Technical specifications	15
2.7	Components and materials used for device construction.....	16
2.8	Mechanism of action/principle of operation.....	21
2.9	Intended use	22
2.10	Indications.....	23
2.11	Contraindications.....	23
2.12	Intended patient population.....	23
2.13	Intended users	23
2.14	Clinical benefits.....	24
2.15	Device claims.....	24
2.16	Instructions for use	25
2.17	Potential complications/adverse events	25
2.18	Manufacturing process	25
2.19	Device Change History	26
3	Clinical background, current knowledge & state-of-art	27
3.1	Identification of pertinent data	27
3.2	Applicable standards, guidelines, and common specifications	28
3.3	Clinical Background.....	31
3.4	Historical Context.....	32
3.5	Side effects and known failure modes/hazards	33
3.6	Alternative Treatment Options.....	34
3.7	Equivalent device	34
3.8	Acceptability of undesired side effects and risks	37
3.9	Conclusion on state-of-the-art analysis	39
4	Device under evaluation	40
4.1	Route of evaluation.....	40
4.2	Justification for WET	44

	BIORAD MEDISYS PVT. LTD.	Document No.	CER/DJS/61
		Revision. No	01
	CLINICAL EVALUATION REPORT PRODUCT- DOUBLE J STENT	Revision Date	26-12-2025

4.2.1	Equivalency Conclusion.....	46
4.2.2	Conformity Assessment Summary.....	47
4.3	Data generated and held by the manufacturer	47
4.3.1	Pre-clinical data.....	48
4.3.1.1	Biocompatibility data	49
4.3.1.2	Bench testing (performance testing).....	52
4.3.2	Clinical data	53
4.3.2.1	Clinical investigations.....	53
4.3.2.2	Post-market surveillance data	53
4.4	Critical appraisal of clinical data	64
4.5.1	Appraisal method and criteria	64
4.5.2	Appraisal of selected articles	67
4.5	Data analysis	75
4.5.1	Requirement on safety	75
4.5.2	Overall Safety and Performance analysis	78
4.5.3	Requirement on acceptable benefit-risk profile.....	80
4.5.4	Requirement on acceptability of side-effects.....	86
5	Conclusion.....	86
6	Date of the next evaluation	87
7	Qualification of responsible evaluators	88
8	References	89
10.	Other references.....	90
	Revision history.....	91
	Document approvals.....	91

	BIORAD MEDISYS PVT. LTD.	Document No.	CER/DJS/61
		Revision. No	01
	CLINICAL EVALUATION REPORT PRODUCT- DOUBLE J STENT	Revision Date	26-12-2025

List of tables

Table 1: Product Scope	11
Table 2: Manufacturer's Details.....	11
Table 3: Details of Person Responsible for Regulatory Compliance (PRRC)	12
Table 4: Regulatory details	12
Table 5: Authorized Representative Details	13
Table 6: Notified Body Details	13
Table 7: Type of assessment	13
Table 8: Technical specifications of the device.....	15
Table 9: Key functional elements of Double J Stent/Kit (Long Term).....	17
Table 10: List of Standards and guidelines	28
Table 11: Alternatives Options – Data from Literature	26
Table 12: Equivalent device	34
Table 13: Analysis of Safety and Performance Outcomes: Equivalent device	37
Table 14: Performance criteria	38
Table 15: Safety Criteria.....	39
Table 16: Comparison of Different Procedures	40
Table 17: Equivalence Table	41
Table 18: Details of Biocompatibility tests	Error! Bookmark not defined.
Table 19: Tests per IEC 60601	53
Table 20: Bench marking study.....	Error! Bookmark not defined.
Table 21: Distribution pattern	Error! Bookmark not defined.
Table 22: Feedback details.....	Error! Bookmark not defined.
Table 23: USFDA Recalls data search	Error! Bookmark not defined.
Table 24: Included Results of Product Recalls	Error! Bookmark not defined.
Table 25: Technical complications reported in TPLC	Error! Bookmark not defined.
Table 26: Search results from MHRA	Error! Bookmark not defined.
Table 27: Search results from TGA	Error! Bookmark not defined.
Table 28: Search results from DHPR	Error! Bookmark not defined.
Table 29: Search results from BfArm	Error! Bookmark not defined.
Table 30: Search results from CDSCO	Error! Bookmark not defined.
Table 31: Cumulative Literature Search Data from all sources ..	Error! Bookmark not defined.
Table 32: Appraisal criteria	65
Table 33: Appraisal scores	66
Table 34: Appraisal Criteria for Suitability and Data Contribution	67
Table 35: Safety criteria of subject device	71
Table 36: Performance criteria of subject device	72

	BIORAD MEDISYS PVT. LTD.	Document No.	CER/DJS/61
		Revision. No	01
	CLINICAL EVALUATION REPORT PRODUCT- DOUBLE J STENT	Revision Date	26-12-2025

Table 37: Parameters for analysis of the benefits77

Table 38: Parameters for analysis of the risks88

Table 39: Qualification of responsible evaluators88

List of figures

Figure 1:14

Figure 2: **Error! Bookmark not defined.**

Figure 3: **Error! Bookmark not defined.**

Figure 4:26

	BIORAD MEDISYS PVT. LTD.	Document No.	CER/DJS/61
		Revision. No	01
	CLINICAL EVALUATION REPORT PRODUCT- DOUBLE J STENT	Revision Date	26-12-2025

List of abbreviation(s)

AE	Adverse Events
BfArm	The Federal Institute for Drugs and Medical Devices (Bundesinstitut für Arzneimittel und Medizinprodukte)
CAPA	Corrective Action and Preventive Action
CE	Conformité Européenne or European Conformity
CEP	Clinical Evaluation Plan
CER	Clinical Evaluation Report
CV	Curriculum Vitae
DHPR	Drug & Health Product Register (Health Canada)
DOI	Declaration of Interest
EMDN	European Medical Device Nomenclature
EPMC	Europe PubMed Central
ETO	Ethylene Oxide
EU	European Union
FDA	Food and Drug Administration
FSCA	Field Safety Corrective Actions
FSN	Field Safety Notice
GSPR	General Safety and Performance Requirements
IFU	Instructions for Use
IMDRF	International Medical Device Regulators Forum
ISO	International Organization for Standardization
LSP	Literature Search Protocol
LSR	Literature Search Report

	BIORAD MEDISYS PVT. LTD.	Document No.	CER/DJS/61
		Revision. No	01
	CLINICAL EVALUATION REPORT PRODUCT- DOUBLE J STENT	Revision Date	26-12-2025

MAUDE	Manufacturer and User Facility Device Experience
MDCG	Medical Device Coordination Group
MDR	Medical Device Regulations
MEDDEV	Medical Devices
MEDLINE	Medical Literature Analysis and Retrieval System Online, or MEDLARS Online
MHRA	Medicines and Healthcare products Regulatory Agency
NB	Notified Body
PCNL	Percutaneous nephrolithotomy
PICO	Patient/Population, Intervention, Comparison and Outcomes
PMCF	Post-Market Clinical Follow-Up
PMS	Post-Market Surveillance
PRISMA	Preferred Reporting Items for Systematic Reviews and Meta-Analyses
QMS	Quality Management System
SoTA	State of-The-Art
SRN	Single Registration Number
TCF	Technical File
TGA	Therapeutic Goods Administration
UK	United Kingdom
US	United States of America
WET	Well Established Technology

	BIORAD MEDISYS PVT. LTD.	Document No.	CER/DJS/61
		Revision. No	01
	CLINICAL EVALUATION REPORT PRODUCT- DOUBLE J STENT	Revision Date	26-12-2025

1 Summary

This Clinical Evaluation Report (CER) has been prepared for the Double J Stent (Short Term), a sterile, single-use ureteral stent intended for internal urinary drainage from the kidney to the bladder. For the purposes of this report and all subsequent sections, the terms DJS are used interchangeably to refer to the same device under evaluation.

This CER has been developed in accordance with the requirements of Regulation (EU) 2017/745 (EU MDR), with the objective of demonstrating conformity with the applicable General Safety and Performance Requirements (GSPRs) set out in Annex I. The clinical evaluation provides a systematic and critical assessment of the safety, clinical performance, and benefit–risk profile of the Double J Stent (Short Term), taking into account its intended purpose, design characteristics, duration of use (short-term: up to 30 days), and defined target patient population (adult male and female patients undergoing urological procedures) as specified by the manufacturer.

As the Double J Stent (Short Term) is a well-established device type within urological practice and clinical investigation data specific to the subject device are limited, the clinical evaluation has been conducted in accordance with Article 61(10) and Annex XIV, Part A of the EU MDR. Accordingly, the evaluation is primarily based on:

- Available device-specific clinical data, including complaint handling records and post-market surveillance information
- Device-specific non-clinical data, including design verification and validation testing, biological safety evaluation in accordance with the ISO 10993 series, sterilization validation, packaging validation, and performance testing as per applicable harmonized standards
- A critical review of relevant scientific literature describing the current state of the art for ureteral stents and short-term internal urinary drainage
- Risk management activities performed in accordance with EN ISO 14971, including identification of known and foreseeable risks and evaluation of the overall benefit–risk balance
- Post-Market Surveillance (PMS) and Post-Market Clinical Follow-Up (PMCF) activities implemented in accordance with Annex III and Annex XIV, Part B of the EU MDR, to monitor residual risks and confirm continued safety and performance throughout the device lifecycle

	BIORAD MEDISYS PVT. LTD.	Document No.	CER/DJS/61
		Revision. No	01
	CLINICAL EVALUATION REPORT PRODUCT- DOUBLE J STENT	Revision Date	26-12-2025

Where applicable, equivalence has been considered in accordance with Article 61 and Annex XIV of the EU MDR, and published clinical literature has been used to characterize the state of the art, expected clinical outcomes, and known risks associated with ureteral stents as a device category.

The indirect and direct clinical evidence collated and assessed within this CER has been generated, appraised, and documented in accordance with Annex XIV, Part A of Regulation (EU) 2017/745 and relevant MDCG guidance documents on clinical evaluation and PMS.

The clinical evaluation is structured to ensure clear traceability to the applicable General Safety and Performance Requirements (GSPRs) specified in Annex I of the EU MDR. Based on the available clinical evidence and post-market data, the evaluation confirms that the Double J Stent (Short Term) demonstrates an acceptable level of safety and clinical performance for its intended purpose. Any identified residual risks or uncertainties are addressed through established PMS and PMCF measures, ensuring ongoing conformity with the EU MDR throughout the device lifecycle

	BIORAD MEDISYS PVT. LTD.	Document No.	CER/DJS/61
		Revision. No	01
	CLINICAL EVALUATION REPORT PRODUCT- DOUBLE J STENT	Revision Date	26-12-2025

1.2 Scope of the clinical evaluation

The European Union (EU) Medical Device Regulation 2017/745 (EU MDR)

The EU MDR aims to ensure a high level of protection of human health and safety and the proper functioning of the EU medical device market. It introduces stricter, more detailed requirements for clinical evaluation. Clinical evaluation is a systematic and ongoing process of collecting, appraising, and analyzing clinical evidence to verify the device's safety, performance, and clinical benefits. It forms a core part of the conformity assessment procedure required for CE marking, demonstrating compliance with the General Safety and Performance Requirements (GSPRs) in Annex I and confirming that the device is suitable for its intended purpose.

Article 61 Requirements

Article 61 outlines the responsibilities of the manufacturer, the notified body (NB), and, where applicable, the expert panel. The manufacturer prepares a Clinical Evaluation Plan (CEP) defining scope, criteria, and methods, and a Clinical Evaluation Report (CER) documenting results and conclusions. The NB assesses both documents, while expert panels provide scientific advice for high-risk or innovative devices. Clinical evaluation must be performed by qualified, impartial personnel and be appropriate for the device's intended purpose, risk class, target population, and expected clinical benefits. The evaluation must rely on sufficient, relevant, and scientifically valid clinical data from clinical investigations, clinical experience, scientific literature, or a combination thereof.

Clinical Evaluation Principles

The clinical evaluation follows a methodologically sound and transparent procedure aligned with MEDDEV 2.7/1 Rev. 4 and applicable MDCG guidance. It includes defining the scope and objectives, identifying and appraising clinical data, generating clinical evidence, and assessing safety, performance, and clinical benefits. The CER documents conclusions, limitations, evidence gaps, and the need for further clinical investigations or PMCF, and is updated throughout the device lifecycle. The manufacturer determines and justifies the level of clinical evidence necessary to demonstrate conformity with the GSPRs.

Use of EU MDR Article 61(10)

	BIORAD MEDISYS PVT. LTD.	Document No.	CER/DJS/61
		Revision. No	01
	CLINICAL EVALUATION REPORT PRODUCT- DOUBLE J STENT	Revision Date	26-12-2025

For certain devices, including accessories, manufacturers may apply Article 61(10) when non-clinical data alone such as performance testing, bench testing, and pre-clinical evaluation are sufficient to demonstrate conformity. This route does not apply to Class III or implantable devices. If clinical claims are made that imply clinical benefit, relevant clinical data must be generated and evaluated, and Article 61(10) no longer applies.

Expert Guidance

Article 61(10) does not exclude available clinical data; rather, it applies when clinical data are not appropriate for demonstrating conformity. Literature screening remains necessary to identify risks and ensure state-of-the-art alignment in accordance with Annex III. The provision is not intended for devices lacking clinical data, but for devices where conformity can be demonstrated using robust non-clinical evidence.

2.1 Identification of device(s)

The scope of this clinical evaluation includes the Double J Stent (Short Term), designed and manufactured by Biorad under the brand name DJS as mentioned below.

Table 1: Product Scope

Common Name	Double J Stent (Short Term)
Trade Names	Double J Stent (Short Term)
Variants	Please see attached annexure for variants

2.2 Name and address of the manufacturer

The details regarding the manufacturer, regulatory compliance personnel, administrative particulars and Notified Body (NB) are presented in the below tables:

Table 2: Manufacturer's Details

Name	BIORAD MEDISYS PVT. LTD.
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	BIORAD MEDISYS PVT. LTD.	Document No.	CER/DJS/61
		Revision. No	01
	CLINICAL EVALUATION REPORT PRODUCT- DOUBLE J STENT	Revision Date	26-12-2025

Address	Survey No.: 48/3 & 48/7, Pashan Sus Road, Sus Village, Taluka Mulshi, Pune- 411021 Maharashtra, INDIA.
Phone	Tel.: +91-20-3091-2000, Fax: +91-20-3091-2050
Email	pune@bioradmedisys.com
Website	www.biorad-medisys.com
Single Registration Number (SRN):	IN-MF-000013050

Table 3: Details of Person Responsible for Regulatory Compliance (PRRC)

Name	Mr. Sudhakar Saxena
Phone	+41 447090101
Email	

2.3 Administrative particulars

Table 4: Regulatory details

Basic UDI-DI:	Basic UDI-DI Codes	
	Double J Stent & Kit Blue (Short Term)	890551639TL
	Blue HC	8905516101AW
	Blue Set	890551640T5
	Blue Set HC	8905516100AV
	Double J Stent & Kit White (Short Term)	890551641T7
	White HC	8905516107BA
	White Set	890551642T9
	White Set HC	8905516108BC

	BIORAD MEDISYS PVT. LTD.	Document No.	CER/DJS/61
		Revision. No	01
	CLINICAL EVALUATION REPORT PRODUCT- DOUBLE J STENT	Revision Date	26-12-2025

QMS Certificate No:	Q5 106138 0001 Rev. 01			
CE Certificate No.				
EMDN Code:	EMDN Codes	MDN Code	MDS Code	MDT Code
	U020302	MDN 1201	MDS 1005	MDT 2002, MDT 2008, MDT 2011
EMDN Code Description:	DEVICES FOR UROGENITAL SYSTEM			
CE approval	Approved			

Table 5: Authorized Representative Details

Name:	CMC Medical Device & Drugs S.L.
SRN:	ES-AR-000000293
Address:	C/ Horacio Lengo N° 18 CP 29006, Malaga- Spain
Email:	Info@cmcmmedicaldevices.es
Phone:	+34951214054

Table 6: Notified Body Details

Name:	TÜV SÜD Product Service GmbH
NB number:	0123
Address:	Ridlerstraße 65 80339 MÜNCHEN, Germany
Email:	ps.zert@tuev-sued.de
Phone:	+49 89 50084747

Table 7: Type of assessment

Initial conformity assessment	☐
Assessment of changes and update of the clinical evaluation	☒
Re-certification assessment	☐

	BIORAD MEDISYS PVT. LTD.	Document No.	CER/DJS/61
		Revision. No	01
	CLINICAL EVALUATION REPORT PRODUCT- DOUBLE J STENT	Revision Date	26-12-2025

2.4 Device description

Double J Stent is used for treatment of the upper urinary tract are implantable devices that support the flow of the urine from the kidneys to the bladder. Double J Stents can be used short-term for up to 30 days, or long-term for several months depending on the cause and course of treatment.

The term Double-J refers to the shape of the stent ends, the purpose of which is to anchor the stent in the kidney and in the bladder to prevent its displacement or to avoid migration, Double J stent contains the Guidewire (PTFE/Hydrophilic coating) which is used in ureteroscopic procedures to access the ureter, identify the ureteral tract and guide instrumentation.

Pusher and guidewire connected as a unit which allows for manipulation or push the stents to accurate position over a compatible Guidewire & suture or string partially inserted guide wire and pusher together stabilize the distal end of the stent and keeps everything adequately aligned with only gentle traction on the dangler thread which exits from the urethra for extra support to avoid the migration. Umbilical cord clamp use to hold the pusher and stents over a guidewire during insertion of stents.

Double J Stents consist of: Stent, Pusher With or without thread(string/suture)

Double J stent Kit Consist Of: Stent, Pusher, Guidewire, Clamps with or without thread(string/suture)

It is supplied as sterile/nonsterile product in a peel-open package and intended for one time use only.

2.5 Device classification

As per the classification criteria set out in Annex VIII of Regulation (EU) 2017/745, the Double J Stent (Short Term) is intended for continuous use for a duration between 60 minutes and up to 30 days, and is therefore considered a short-term use device.

The risk class of the device and the justification for the applicable classification rule have been determined in accordance with Annex VIII of the EU MDR. The Double J Stent (Short Term) is an invasive device introduced into the body via a natural orifice (urethra) and is intended for internal urinary drainage from the kidney to the bladder.

	BIORAD MEDISYS PVT. LTD.	Document No.	CER/DJS/61
		Revision. No	01
	CLINICAL EVALUATION REPORT PRODUCT- DOUBLE J STENT	Revision Date	26-12-2025

In accordance with Rule 5 of Annex VIII, invasive devices with respect to body orifices intended for short-term use are classified as Class IIa. Therefore, the Double J Stent (Short Term) is classified as a Class IIa medical device under Regulation (EU) 2017/745.

2.6 Technical specifications

Table 8: Technical specifications of the Double J Stent (Short Term)

Parameter	Specifications
Design	Double J (pigtail at both ends) ureteral stent for internal urinary drainage from kidney to bladder
Material	Stent: Polyurethane (White/Blue) Pusher: Polypropylene String/Suture: Polypropylene Guidewire (in kit): SS 304 with PTFE coating / Nitinol with hydrophilic coating
Available Sizes (Diameter)	Multiple variants available (refer product variant list in Technical File)
Available Lengths	Multiple length options available (refer product variant list in Technical File)
Radiopacity	Radiopaque (device visualization under imaging)
Tip Configuration	Tapered ends for ease of introduction over compatible guidewire; bilateral pigtail (Double J) configuration
Working Channel Compatibility	Compatible with standard urological guidewires used in ureteroscopic procedures
Retention Mechanism	Bilateral pigtail (Double J) ends to prevent migration; optional polypropylene string for distal stabilization
MRI Compatibility	Not specified in Technical File (no MR safety claim declared)
Sterilization Method	Ethylene Oxide (EtO) sterilized

	BIORAD MEDISYS PVT. LTD.	Document No.	CER/DJS/61
		Revision. No	01
	CLINICAL EVALUATION REPORT PRODUCT- DOUBLE J STENT	Revision Date	26-12-2025

Parameter	Specifications
Packaging	Primary: Polyurethane cover with Tyvek pouch Secondary: Duplex box Tertiary: 5-ply corrugated box
Shelf Life	5 years
Color Coding	Available in Blue and White variants
Insertion Method	Introduced via natural orifice (urethra) over compatible guidewire using pusher
Stent Pusher Included	Yes (Pusher included; available with/without radiopaque tip; kit version includes guidewire and clamp)
Pigtail Memory Retention	Designed to maintain pigtail configuration to reduce risk of migration
Flow Holes	Multiple drainage openings along stent body to facilitate urine flow
Drainage Type	Internal urinary drainage from kidney to bladder
Removal Options	Endoscopic removal; optional string-assisted removal
Use Duration	Short-term use (60 minutes up to 30 days)
Biocompatibility	Evaluated in accordance with ISO 10993 series; passed Cytotoxicity, Intracutaneous Reactivity, and Skin Sensitization testing

2.7 Components and materials used for device construction

The Double J Stent (Short Term) consists of the following components and materials, as defined in the TCF/DJS/61:

Component	Material	Type of Contact	Function
Double J Stent (Short Term)	Polyurethane (White / Blue)	Direct	Provides internal urinary drainage from kidney to bladder; maintains patency of ureter

	BIORAD MEDISYS PVT. LTD.	Document No.	CER/DJS/61
		Revision. No	01
	CLINICAL EVALUATION REPORT PRODUCT- DOUBLE J STENT	Revision Date	26-12-2025

Component	Material	Type of Contact	Function
String (optional)	Polypropylene	Direct	Assists in distal stabilization and facilitates stent removal
String Sheath	LDPE (Blue)	Indirect	Secures the polypropylene string

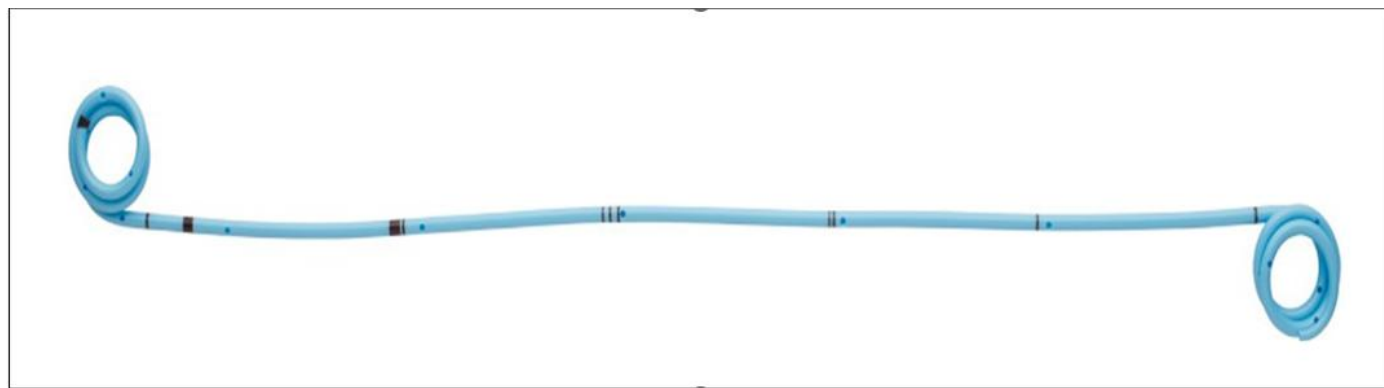
Component	Material	Type of Contact	Function
Pusher / Introducer Sheath	Polypropylene (Red/Brown/Green/Blue)	Direct	Used to position and deploy the stent accurately over a guidewire
Guidewire	Stainless Steel 304 with PTFE coating / Nitinol with Hydrophilic coating	Direct	Facilitates access to the ureter and guides stent placement
Clamp (Umbilical Cord Clamp)	Polypropylene	Indirect	Holds the stent and pusher in position during insertion

The materials selected for device construction are widely used in urological medical devices and are suitable for short-term internal use. The device materials have undergone biological safety evaluation in accordance with the ISO 10993 series to ensure biocompatibility for their intended duration and type of contact.

The combination of polyurethane for the stent body and polypropylene-based accessory components ensures flexibility, mechanical strength, chemical resistance, and compatibility with Ethylene Oxide (EtO) sterilization, thereby supporting the safety and performance of the Double J Stent (Short Term) for its intended use.

Table 9: Key functional elements of Double J Stent (Short Term)

	BIORAD MEDISYS PVT. LTD.	Document No.	CER/DJS/61
		Revision. No	01
	CLINICAL EVALUATION REPORT PRODUCT- DOUBLE J STENT	Revision Date	26-12-2025



Double J Stent (Short Term)

Product Codes	Product Description	Intended Use	Type of Use	Material
Blue HC	Sterile, single-use Double J (bilateral pigtail) ureteral stent supplied with/without string; kit version includes pusher, guidewire, and clamp	Intended for internal urinary drainage from kidney to bladder in conditions such as ureteral obstruction, calculi, hydronephrosis, ureteral stricture, and post-urological procedures	Short-term use (60 minutes up to 30 days); Single-use; Invasive via natural orifice	Stent: Polyurethane (Blue) Pusher: Polypropylene String: Polypropylene Guidewire (kit): SS 304 with PTFE coating / Nitinol with hydrophilic coating
White Set HC	Sterile, single-use Double J (bilateral pigtail) ureteral stent supplied with/without string; kit version includes pusher, guidewire, and clamp	Intended for internal urinary drainage from kidney to bladder in conditions such as ureteral obstruction, calculi, hydronephrosis, ureteral stricture, and post-urological procedures	Short-term use (60 minutes up to 30 days); Single-use; Invasive via natural orifice	Stent: Polyurethane (White) Pusher: Polypropylene String: Polypropylene Guidewire (kit): SS 304 with PTFE coating / Nitinol with hydrophilic coating

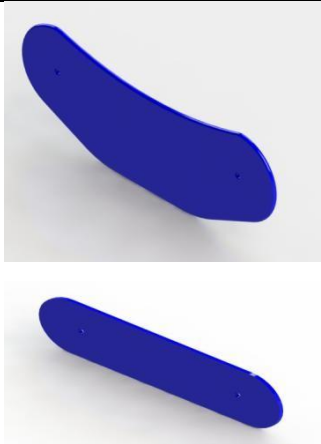
● **PART FUNCTIONS**

PART	FUNCTION	IMAGE
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	BIORAD MEDISYS PVT. LTD.	Document No.	CER/DJS/61
		Revision. No	01
	CLINICAL EVALUATION REPORT PRODUCT- DOUBLE J STENT	Revision Date	26-12-2025

Double J Stents - Short Term	• Devices that support the flow of the urine from the kidneys to the bladder. • These are made available with tapered ends to allow for ease of introduction over a compatible guidewire. Further its smooth surface also provides for lesser trauma for the patients. Device comprises pigtail shape to avoid migration. • Short term: Double J Stents is normally intended for continuous use for between 60 minutes and 30 days	
String	Polypropylene suture or thread is used for stabilizing the distal end of the stent and keeps everything adequately aligned with only gentle traction on the dangler thread which exits from the urethra for extra support to avoid the migration or ease	

	BIORAD MEDISYS PVT. LTD.	Document No.	CER/DJS/61
		Revision. No	01
	CLINICAL EVALUATION REPORT PRODUCT- DOUBLE J STENT	Revision Date	26-12-2025

	repositioning and stent removal	
String sheath	Securely hold the Polypropylene thread.	

Accessories

PART	FUNCTION	IMAGE
Umbilical cord clamp	Umbilical cord clamp use to clamp or hold the pusher and Double J Stents or ureteral stents during insertion of stents over a guidewire (all category)	

	BIORAD MEDISYS PVT. LTD.	Document No.	CER/DJS/61
		Revision. No	01
	CLINICAL EVALUATION REPORT PRODUCT- DOUBLE J STENT	Revision Date	26-12-2025

Pusher	Pusher (introducer sheath) with/without Radiopaque pusher and guidewire connected as a unit which allows for manipulation or push the stents to accurate position over a compatible guidewire (all category) Pusher available in following type 1. Pusher 2. Pusher with Radiopaque tip 3. Locking Pusher	
Guidewire (all category)	1. Stiff proximal shaft ensures push ability and control. 2. Floppy distal tip enables precise, less traumatic advancement. 3. Smooth, rounded ends and PTFE coating enables easy passage 4. Stainless steel-304 with PTFE	

2.8 Mechanism of action/principle of operation

The Double J Stent (Short Term) is a passive, implantable ureteral stent intended to support urine flow from the kidney to the bladder in cases of ureteral obstruction or impaired drainage.

The mechanism of action is mechanical and is based on the following:

	BIORAD MEDISYS PVT. LTD.	Document No.	CER/DJS/61
		Revision. No	01
	CLINICAL EVALUATION REPORT PRODUCT- DOUBLE J STENT	Revision Date	26-12-2025

1. Internal Urinary Drainage
The stent is placed within the ureter, extending from the renal pelvis to the urinary bladder. Its hollow tubular structure provides a conduit that allows urine to pass from the kidney to the bladder when normal ureteral flow is compromised.
2. Bypassing Ureteral Obstruction
In medical conditions such as calculi, hydronephrosis, ureteral stricture, or post-procedural swelling, the stent maintains patency of the ureter by creating an open channel for urine passage.
3. Anchoring via Double J (Pigtail) Configuration
The proximal and distal ends are shaped into pigtails (“Double J”) to anchor the device within the renal pelvis and bladder, thereby reducing the risk of migration.
4. Guidewire-Assisted Placement
The stent is introduced over a compatible guidewire using a pusher system, allowing accurate positioning within the ureter.
5. Temporary Support Function
The device is intended for short-term use (up to 30 days) and provides temporary mechanical support to ensure drainage until the underlying condition is resolved or further intervention is performed.

Principle of Operation and Mode of Action

Double J Stent is a thin, flexible plastic tube that is placed temporarily in the ureter to help urine drain from the kidney into the bladder in the case of a blockage.

2.9 Intended use

This is a single use sterile device which is used for internal drainage from Kidney to Bladder in medical conditions like improper urine drainage, calculi in ureter, hydronephrosis, oxalate calculi, pyeloplasty with UPJ Obstruction, residual renal calculus, ureteral stricture.

	BIORAD MEDISYS PVT. LTD.	Document No.	CER/DJS/61
		Revision. No	01
	CLINICAL EVALUATION REPORT PRODUCT- DOUBLE J STENT	Revision Date	26-12-2025

2.10 Indications

This is a single use sterile device used for internal drainage from Kidney to Bladder

2.11 Contraindications

There are no known absolute contraindications for this device.

2.12 Intended patient population

Double J Stent (Short term) is manufactured to meet requirements of medical conditions like Improper urine drainage, calculi in ureter, hydronephrosis, oxalate calculi, pyeloplasty with UPJ Obstruction, residual renal calculus, ureteral stricture. These medical conditions commonly occur in adult Male, Female during urological procedures. Hence Double J Stent (Short term) can be use in Adults, Male / Female.

2.13 Intended users

The Double J Stent (Short Term) is intended to be used by qualified healthcare professionals trained and experienced in urological procedures.

The intended users include:

- Urologists
- Surgeons trained in endourology or urological interventions
- Interventional specialists performing ureteroscopic procedures
- Trained operating room or catheterization laboratory personnel under physician supervision

The device is intended for use in a clinical/hospital setting, including operating theatres, endoscopy suites, and other controlled healthcare environments. Proper knowledge of ureteral anatomy, guidewire-assisted placement techniques, and stent positioning is required for safe and effective use.

	BIORAD MEDISYS PVT. LTD.	Document No.	CER/DJS/61
		Revision. No	01
	CLINICAL EVALUATION REPORT PRODUCT- DOUBLE J STENT	Revision Date	26-12-2025

2.14 Clinical benefits

The Double J Stent (Short Term) is intended for internal urinary drainage from the kidney to the bladder in medical conditions such as ureteral obstruction, calculi, hydronephrosis, ureteral stricture, and during or after urological procedures.

When used as intended, the device provides the following clinical benefits:

1. Maintenance of Ureteral Patency
Provides a temporary internal conduit within the ureter to maintain patency and allow urine to pass from the kidney to the bladder.
2. Restoration of Urine Flow
Supports continuous urinary drainage in cases where normal ureteral flow is compromised.
3. Relief of Obstruction-Related Symptoms
Assists in managing conditions associated with impaired urinary drainage, such as hydronephrosis and obstruction due to calculi or strictures.
4. Support Following Urological Procedures
Maintains drainage and ureteral alignment during the post-procedural healing phase following endourological interventions.
5. Temporary Short-Term Drainage Solution
Provides short-term (up to 30 days) internal drainage support until the underlying condition is resolved or further definitive treatment is performed.

The clinical benefits are achieved through mechanical maintenance of ureteral patency and are consistent with the established clinical practice for short-term ureteral stenting.

2.15 Device claims

The Double J Stent (Short Term) is a sterile, single-use ureteral stent intended for short-term internal urinary drainage from the kidney to the bladder in conditions such as ureteral obstruction, calculi, hydronephrosis, ureteral stricture, pyeloplasty with UPJ obstruction, and residual renal calculi. The device is designed to maintain ureteral patency and support continuous urine flow for a duration of up to 30 days. It features a bilateral pigtail (Double J) configuration to reduce the risk of migration, tapered ends to facilitate guidewire-

	BIORAD MEDISYS PVT. LTD.	Document No.	CER/DJS/61
		Revision. No	01
	CLINICAL EVALUATION REPORT PRODUCT- DOUBLE J STENT	Revision Date	26-12-2025

assisted placement, and multiple drainage openings along the shaft to enable effective urine passage. The device is radiopaque to allow visualization during placement. It is supplied sterile, intended for single use only, and is available with or without a polypropylene string to facilitate stabilization and removal. Kit variants include a pusher, guidewire, and clamp to assist in controlled insertion. The materials used are suitable for short-term internal use and have undergone biological safety evaluation, and the device is sterilized using a validated Ethylene Oxide (EtO) process. The Double J Stent (Short Term) is classified as a Class IIa medical device under Regulation (EU) 2017/745.

2.16 Instructions for use

Refer IFU Document No: BRP/IFU/DJS/01/03

2.17 Potential complications/adverse events

The use of the Double J Stent (Short Term), as with other ureteral stents, may be associated with certain foreseeable complications and adverse events. Based on the device description, intended use, and known risks associated with ureteral stenting procedures, the potential complications include:

- Urinary tract infection (UTI)
- Stent migration or malposition
- Hematuria

2.18 Manufacturing process

The manufacturing process of the **Double J Stent/Kit (Short Term)** begins with inspection of semi-finished components to ensure compliance with predefined specifications. Approved components undergo shaping, hole punching, tapering, marking, cleaning, and assembly processes, including guidewire integration for kit variants. In-process inspections are performed at key stages, and non-conforming parts are rejected. Following final inspection, the product is packaged, labeled, and sterilized using validated Ethylene Oxide

	BIORAD MEDISYS PVT. LTD.	Document No.	CER/DJS/61
		Revision. No	01
	CLINICAL EVALUATION REPORT PRODUCT- DOUBLE J STENT	Revision Date	26-12-2025

(EtO) methods. After sterilization and quality review, the finished goods are released for distribution under the established Quality Management System.

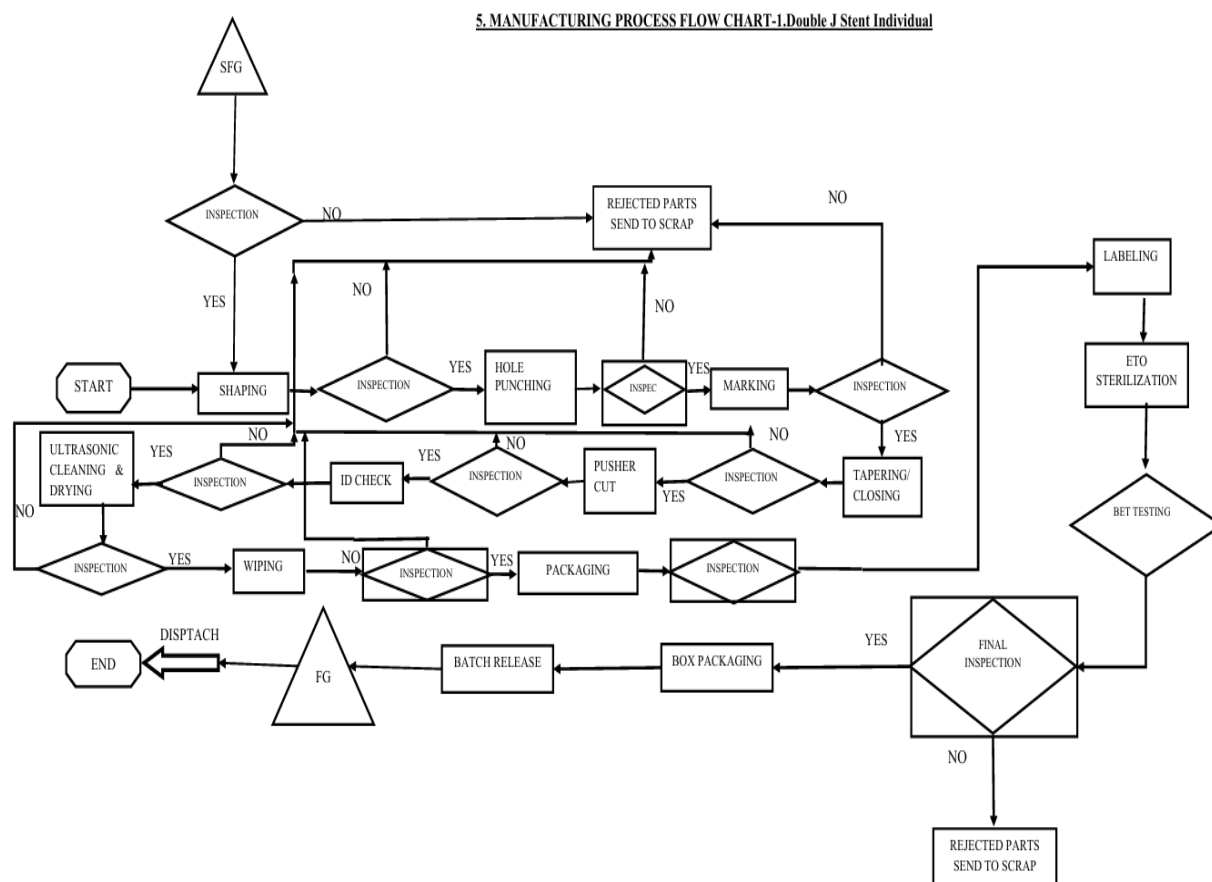


Figure 1

2.19 Device Change History

The Double J Stent/Kit (Long Term) is a first-generation device. However, similar generic devices have been on the market for some time and have well-established technology (WET)

	BIORAD MEDISYS PVT. LTD.	Document No.	CER/DJS/61
		Revision. No	01
	CLINICAL EVALUATION REPORT PRODUCT- DOUBLE J STENT	Revision Date	26-12-2025

3 Clinical background, current knowledge & state-of-art

In the context of medical devices, "state-of-the-art" refers to the developed stage of current technical capability and/or accepted clinical practice in relation to products, processes, and patient management. This concept, as outlined in the MDCG (Medical Device Coordination Group) 2020-6 guidelines, reflects the consensus of scientific, technological, and clinical experience at a given time. It signifies the standard of what is widely recognised and implemented as good practice within the fields of technology and medicine. However, it is important to note that the state-of-the-art does not necessarily correspond to the most advanced or cutting-edge solutions, but rather to those that are widely accepted and demonstrate proven effectiveness in clinical use.

The following are discussed in this section:

- Clinical background to identify relevant medical conditions
- Stages of medical conditions in terms of severity, rate of occurrence, affected populations
- Available medical alternatives for relevant medical conditions
- Advantages and disadvantages of medical alternatives in comparison to the device under evaluation
- Historical context and development of the device under evaluation
- Safety and performance characteristics of the device under evaluation
- Applicable regulatory references for the device under evaluation

3.1 Identification of pertinent data

A literature search was conducted to discover and compile literature that describes available management options, their advantages and disadvantages, their benefit/risk profiles, and limitations and which provides sufficient information for comparison with state-of-the art.

Non-systematic literature review

A non-systematic literature review is an informal and targeted review of published scientific information conducted to support the clinical evaluation of the device. Unlike a systematic literature review, it does not follow a fully pre-defined protocol for comprehensive searching, study selection, and critical appraisal, and therefore may not capture all available evidence. For the Double J Stent (Short Term), a non-systematic literature review may be used to obtain background information on the state of the art, expected clinical

	BIORAD MEDISYS PVT. LTD.	Document No.	CER/DJS/61
		Revision. No	01
	CLINICAL EVALUATION REPORT PRODUCT- DOUBLE J STENT	Revision Date	26-12-2025

outcomes, and known risks/complications associated with ureteral stents as a device category. The information identified through this approach is used primarily for contextual understanding and to support clinical rationale; it is not relied upon as the sole basis for demonstrating conformity and does not replace the requirement for a systematic and critical assessment of clinical evidence under the EU MDR.

Systematic literature review

In addition to the initial non-systematic search, relevant literature through a systematic search was also conducted. Following sources were used for the collection of relevant information.

1. PubMed (<https://pubmed.ncbi.nlm.nih.gov/advanced/>)
2. Cochrane (<https://www.cochranelibrary.com/advanced-search>)
3. ClinicalTrials.gov (<https://www.clinicaltrials.gov/>)
4. EPMC (<https://europepmc.org/>)

Literature search was conducted to gather sufficient information on the following research questions:

- What clinical conditions can the Double J Stent (Short term) be used in?
- What are the alternatives for the diagnosis and treatment of conditions affecting the ureters and kidneys?
- Are there any similar Double J Stent (Short term) marketed by other manufacturers?
- What are the safety and performance outcomes of the Double J Stent (Short term)?
- What are the side-effects of using the Double J Stent (Short term)?

For further details, refer to the search in Appendix I - Literature Search Protocol and Appendix II - Literature Search Report.

3.2 Applicable standards, guidelines, and common specifications

The following standards and guidance documents apply to Double J Stent (Short term), designed and manufactured by Biorad. As presented in Table 10.

Table 10: List of Standards and guidelines

Name of Standard / Guideline	Latest Version
Quality management systems – Requirements	ISO 9001:2015

	BIORAD MEDISYS PVT. LTD.	Document No.	CER/DJS/61
		Revision. No	01
	CLINICAL EVALUATION REPORT PRODUCT- DOUBLE J STENT	Revision Date	26-12-2025

Medical devices — Quality management systems — Requirements for regulatory purposes	EN ISO 13485:2016 + A11:2021
Medical devices — Application of risk management to medical devices	EN ISO 14971:2019 + A11:2021
Clean rooms and associated controlled environments – Part 1	EN ISO 14644-1:2015
Clean rooms and associated controlled environments – Part 2	EN ISO 14644-2:2015
Clean rooms and associated controlled environments – Part 3	EN ISO 14644-3:2019
Sterilization of medical devices — Requirements for devices to be designated “STERILE”	EN 556-1:2021
Sterilization of health-care products — Ethylene oxide	ISO 11135:2014 + A1:2018
Sterilization of health care products — Microbiological methods – Part 1	ISO 11737-1:2018 + A1:2021
Sterilization of medical devices — Microbiological methods – Part 2	EN ISO 11737-2:2019
Sterilization of health care products — Biological indicators – Part 2	EN ISO 11138-2:2017
Biological evaluation of medical devices – Part 1	EN ISO 10993-1:2018
Biological evaluation – Cytotoxicity	EN ISO 10993-5:2009
Biological evaluation – EO Residuals	ISO 10993-7:2008 + A1:2019
Biological evaluation – Irritation & Sensitization	EN ISO 10993-10:2021
Biological evaluation – Sample preparation	EN ISO 10993-12:2021
Biological evaluation – Toxicological risk assessment	EN ISO 10993-17:2023
Biological evaluation – Chemical characterization	ISO 10993-18:2020 + A1:2022
Biological evaluation – Tests for irritation	EN ISO 10993-23:2021
Sampling procedures for inspection by variables – Part 1	ISO 3951-1:2022
Sampling procedures for inspection by variables – Part 2	ISO 3951-2:2013
Symbols to be used with medical device labels	EN ISO 15223-1:2021
Information supplied by the manufacturer	EN ISO 20417:2021

	BIORAD MEDISYS PVT. LTD.	Document No.	CER/DJS/61
		Revision. No	01
	CLINICAL EVALUATION REPORT PRODUCT- DOUBLE J STENT	Revision Date	26-12-2025

Packaging for terminally sterilized medical devices – Part 1	EN ISO 11607-1:2019 + A1:2023
Packaging for terminally sterilized medical devices – Part 2	EN ISO 11607-2:2019 + A1:2023
Guidance on application of ISO 11607	ISO/TS 16775:2021
Paper – Determination of bursting strength	ISO 2758:2014
Sealable pouches and reels – Requirements and test methods	EN 868-5:2018
Seal integrity – Visual inspection	ASTM F1886/F1886M-16
Gross leak detection (Bubble Test)	ASTM F2096-19
Leak detection by bubble emission	ASTM D3078-21
Dye penetration seal test	ASTM F1929-15
Seal strength of flexible barrier materials	ASTM F88/F88M-21
Internal pressurization failure resistance	ASTM F1140/F1140M-20
Accelerated aging of sterile barrier systems	ASTM F1980-21
Performance testing of shipping containers	ASTM D4169-22
Tear-propagation resistance of plastic film	ASTM D1938-19
Tensile properties of plastics	ASTM D638-22
Tensile properties of thin plastic sheeting	ASTM D882-18
Application of usability engineering to medical devices	IEC 62366-1:2015 + A1:2020
US Pharmacopoeia	USP–NF (Current Edition)
Post-Market Surveillance Guidance	MDCG 2023-3 (latest PMS guidance)
Clinical Evaluation Requirements	MDR 2017/745 Annex XIV
Clinical Evaluation – Equivalence	MDCG 2020-5
Clinical Evaluation Report	MDCG 2020-13

	BIORAD MEDISYS PVT. LTD.	Document No.	CER/DJS/61
		Revision. No	01
	CLINICAL EVALUATION REPORT PRODUCT- DOUBLE J STENT	Revision Date	26-12-2025

	EU MDR 2017/745
PMCF Plan Template	MDCG 2020-7
PMCF Evaluation Report	MDCG 2020-8
Periodic Safety Update Report (PSUR)	MDCG 2022-21

3.3 Clinical Background

The included scientific articles collectively describe the medical condition addressed by the device, outlining its pathophysiology, clinical presentation, and epidemiological relevance. The literature consistently reflects current clinical practice and state-of-the-art management approaches, detailing standard treatment pathways, commonly used technologies, and their known limitations. The articles highlight key unmet clinical needs, including risks, complications, or performance constraints associated with existing therapeutic options, establishing the rationale for improved or alternative solutions. Evidence from the included publications also defines the typical target patient population, relevant comorbidities, and expected clinical outcomes associated with standard care. This clinical background provides a comprehensive, literature-supported context for assessing the device's intended purpose, anticipated clinical benefits, and comparison against established therapeutic benchmarks within the current state of the art.

Ureteral stents (commonly “Double J” stents) are routinely used in urological practice to maintain ureteral patency and support internal urinary drainage, particularly in the context of ureteral obstruction and peri-procedural management of urinary stone disease. Published evidence consistently shows that, while ureteral stents are clinically useful, stent-related morbidity is common, with urinary symptoms and pain affecting daily activities and health-related quality of life in a substantial proportion of patients; early work by Joshi et al. reported that stent symptoms can reduce functional capacity and patient utility/quality-of-life measures, highlighting the importance of patient counselling and careful stent indication. To enable standardized assessment of these patient-reported outcomes, the Ureteral Stent Symptom Questionnaire (USSQ) was developed and validated as a multidimensional instrument for evaluating stent-related symptoms and quality of life, and has since undergone linguistic validation for broader clinical and research use (e.g., Italian and French versions). Systematic reviews and meta-analyses examining stenting versus no stenting after uncomplicated ureteroscopy/ureteroscopic lithotripsy indicate that routine postoperative stenting does not necessarily improve key effectiveness outcomes (e.g., stone-free results) in uncomplicated cases, while stent-related symptoms remain a relevant burden; therefore, evidence supports a risk-based, indication-driven approach to postoperative stent placement rather than automatic routine use in all patients. In parallel, literature describing

	BIORAD MEDISYS PVT. LTD.	Document No.	CER/DJS/61
		Revision. No	01
	CLINICAL EVALUATION REPORT PRODUCT- DOUBLE J STENT	Revision Date	26-12-2025

ureteroscopy-related ureteral injury has introduced structured assessment tools such as the Post-Ureteroscopic Lesion Scale (PULS) to grade ureteral lesions and improve consistency in reporting, which is relevant to understanding the procedural context in which stents are commonly placed.

3.4 Historical Context

The historical context summarizes the evolution of the medical condition's management and the development of technologies historically used to address it. Published literature describes how clinical practice, treatment methods, and device technologies have progressed over time, including previously established approaches, their limitations, and the shift toward more advanced or safer solutions. The historical evidence highlights the maturity of the therapeutic area, demonstrating how clinical understanding and device performance expectations have evolved to align with the current state of the art. This context supports the justification for the device's intended use by showing how the technology fits within the established clinical landscape and how it addresses limitations identified in earlier generations of treatment or device technologies.

Ureteral stenting has evolved as a fundamental component of modern urological practice, particularly in the management of ureteral obstruction and peri-procedural urinary drainage. The concept of internal ureteral drainage emerged in the late 1960s and 1970s with the development of indwelling ureteral stents designed to maintain ureteral patency without the need for external drainage systems. The introduction of the Double J (pigtail) configuration, featuring curled proximal and distal ends, represented a major advancement by reducing stent migration and improving positional stability within the renal pelvis and bladder. This design has since become the standard configuration for ureteral stents worldwide.

With the expansion of endourological techniques, particularly ureteroscopy and ureteroscopic lithotripsy for stone management, ureteral stents became widely used for postoperative drainage, prevention of obstruction due to edema, and facilitation of ureteral healing. Over time, clinical research shifted from establishing basic effectiveness to evaluating patient-reported outcomes, symptom burden, and quality of life associated with indwelling stents. Studies such as those by Joshi et al. (2003) highlighted the impact of stent-related symptoms, leading to the development of validated instruments such as the Ureteral Stent Symptom Questionnaire (USSQ) to systematically assess stent morbidity.

Subsequent systematic reviews and meta-analyses examined the necessity of routine stenting following uncomplicated ureteroscopy, contributing to a more evidence-based, indication-driven approach to stent placement. This body of literature established the current understanding that ureteral stents provide clear clinical benefits in maintaining ureteral patency and supporting drainage, while also being associated with predictable and manageable side effects.

Today, Double J stents remain the standard of care for short-term internal urinary drainage, with well-characterized performance, safety profiles, and recognized clinical applications in urology.

	BIORAD MEDISYS PVT. LTD.	Document No.	CER/DJS/61
		Revision. No	01
	CLINICAL EVALUATION REPORT PRODUCT- DOUBLE J STENT	Revision Date	26-12-2025

3.5 Side effects and known failure modes/hazards

The Double J Stent (Short Term) is associated with known and foreseeable risks consistent with ureteral stenting procedures and the state of the art.

- Urinary tract infection (UTI): May occur due to the presence of an indwelling device within the urinary tract.
- Hematuria: Mild and usually transient bleeding may occur following placement.
- Stent migration or malposition: Displacement may occur if not positioned correctly.

These side effects and hazards are well-recognized for short-term ureteral stents and are considered manageable when the device is used according to the manufacturer's instructions and within the intended duration of use.

	BIORAD MEDISYS PVT. LTD.	Document No.	CER/DJS/61
		Revision. No	01
	CLINICAL EVALUATION REPORT PRODUCT- DOUBLE J STENT	Revision Date	26-12-2025

3.6 Alternative Treatment Options

The following table provides a comparison of the Double J Stent (Short Term) with alternative treatment options available for the management of ureteral obstruction and impaired urinary drainage. The comparison is based on the intended purpose, expected clinical benefits, and known limitations of each approach.

- **Table 11: Comparison with Alternative Treatment**

Category	Procedures	Clinical Benefits	Risks / Limitations
Internal Ureteral Drainage	Double J Stent (Short Term)	Maintains ureteral patency and enables internal drainage from kidney to bladder; minimally invasive; avoids external collection system; suitable for short-term use following obstruction or urological procedures	Stent-related symptoms, hematuria, risk of infection, migration or encrustation if used beyond intended duration; requires removal
External Urinary Diversion	Percutaneous Nephrostomy (PCN)	Effective decompression of obstructed kidney when retrograde access is not feasible; allows direct renal drainage	Requires external tube and collection bag; risk of infection, bleeding, tube dislodgement; reduced patient comfort and mobility
No Post-Procedural Stenting	Observation after uncomplicated ureteroscopy	Avoids stent-related symptoms; improved short-term patient comfort in carefully selected cases	Not suitable for all patients; risk of postoperative obstruction, pain, or unplanned intervention
Definitive Surgical / Endourological Treatment	Ureteroscopy with stone removal, lithotripsy, stricture management	Addresses underlying cause of obstruction; may restore normal urine flow	Procedural risks; temporary stenting may still be required post-procedure

3.7 Equivalent device

There are many devices commercially available that are similar in technology, mechanism of action, and intended use to the Double J Stent/Kit (Short Term). In this clinical evaluation

	BIORAD MEDISYS PVT. LTD.	Document No.	CER/DJS/61
		Revision. No	01
	CLINICAL EVALUATION REPORT PRODUCT- DOUBLE J STENT	Revision Date	26-12-2025

report, the Black Silicone Filiform Double Pigtail Ureteral Stent (Cook Medical) is considered as the equivalent device for evaluation.

Table 12: Equivalent device

Manufacturer	Device Name	Description	Indications	Regulatory approval status
Cook Medical	Black Silicone Filiform Double Pigtail Ureteral Stent	The Black Silicone Filiform Double Pigtail Ureteral Stent is Used for temporary internal drainage from the ureteropelvic junction to the bladder. The filiform stent assembly provides one-pass placement. The set includes a stent with a tether and stent positioner. This device is not intended to remain	The Black Silicone Filiform Double Pigtail Ureteral Stent is indicated for temporary internal drainage from the ureteropelvic junction to the bladder. Its primary use is to maintain urinary flow when normal passage is obstructed, such as in cases of ureteral blockage due to stones, strictures, malignancy, or other conditions causing partial or complete obstruction. The stent is not intended for permanent implantation and should not remain in place for more than twelve months.	CE Marked

	BIORAD MEDISYS PVT. LTD.	Document No.	CER/DJS/61
		Revision. No	01
	CLINICAL EVALUATION REPORT PRODUCT- DOUBLE J STENT	Revision Date	26-12-2025

		indwelling for more than twelve months.		
--	--	--	--	--

The device Black Silicone Filiform Double Pigtail Ureteral Stent is considered equivalent because it shares the same intended purpose, design features, materials, and mode of action as the subject device. Comprehensive comparisons have been conducted to confirm that the clinical, technical, and biological characteristics are sufficiently similar, ensuring that existing clinical data from the equivalent device can be reliably applied.

	BIORAD MEDISYS PVT. LTD.	Document No.	CER/DJS/61
		Revision. No	01
	CLINICAL EVALUATION REPORT PRODUCT- DOUBLE J STENT	Revision Date	26-12-2025

3.8 Acceptability of undesired side effects and risks

Based on the review of the Technical Documentation, risk management file, and available clinical evidence, the undesired side effects and residual risks associated with the Double J Stent (Short Term) are consistent with the established state of the art for ureteral stents intended for short-term internal urinary drainage.

Identified side effects such as urinary tract infection, hematuria, and potential stent migration are well-recognized and are inherent to ureteral stenting procedures. These risks have been addressed through design features (bilateral pigtail configuration to reduce migration), material selection suitable for short-term indwelling use, validated sterilization processes, and clear labeling and instructions for use. The device is intended for short-term use (up to 30 days), which further mitigates the likelihood of complications such as encrustation or blockage.

Risk management activities conducted in accordance with applicable standards confirm that risks have been reduced as far as possible and that residual risks remain acceptable when weighed against the clinical benefits of maintaining ureteral patency and ensuring effective urinary drainage. Post-market surveillance data, as documented in the Technical File, do not indicate any unexpected safety concerns or trends that would alter the established benefit–risk profile.

Accordingly, the identified undesired side effects and risks are considered acceptable in relation to the intended purpose and clinical benefits of the Double J Stent (Short Term).

Table 13: Side Effects & Risks Reported in Literature

Side Effect	Reported rate range in literature (stented arms)	IFU / risk control linkage from DJS Short Term TCF
Urinary frequency/urgency	58%	Managed via appropriate use per IFU (directions for use, precautions) and clinical monitoring, consistent with the device being supplied with IFU content intended to support safe use.
Haematuria	~19% to 100%	Managed via appropriate use per IFU (directions for use, precautions) and clinical monitoring, consistent with the device being

	BIORAD MEDISYS PVT. LTD.	Document No.	CER/DJS/61
		Revision. No	01
	CLINICAL EVALUATION REPORT PRODUCT- DOUBLE J STENT	Revision Date	26-12-2025

		supplied with IFU content intended to support safe use.
Stent migration / malposition	0.6% – 4%	Managed via appropriate use per IFU (directions for use, precautions) and clinical monitoring, consistent with the device being supplied with IFU content intended to support safe use.

	BIORAD MEDISYS PVT. LTD.	Document No.	CER/DJS/61
		Revision. No	01
	CLINICAL EVALUATION REPORT PRODUCT- DOUBLE J STENT	Revision Date	26-12-2025

3.9 Conclusion on state-of-the-art analysis

The review of the current scientific literature and clinical practice guidelines confirms that ureteral stenting, particularly using Double J–configured stents, represents an established and widely accepted method for short-term internal urinary drainage in patients with ureteral obstruction or following urological procedures. The Double J design, characterized by bilateral pigtail ends to reduce migration and a hollow lumen to maintain ureteral patency, is recognized as the standard configuration in contemporary urological practice.

The analyzed literature demonstrates that ureteral stents are effective in restoring and maintaining urinary drainage and in supporting post-procedural ureteral healing. Reported side effects, including urinary tract infection, hematuria, irritative urinary symptoms, and occasional migration, are well characterized and consistent across studies. These effects are generally transient or manageable and are considered inherent to the device category rather than indicative of device malfunction when used as intended.

The state-of-the-art review also indicates that while alternative management strategies exist (e.g., percutaneous nephrostomy or selective non-stenting in uncomplicated cases), internal ureteral stenting remains an appropriate and clinically justified intervention in indicated patients.

Based on the available evidence, the Double J Stent (Short Term) aligns with the current state of the art in terms of design principles, intended use, safety profile, and clinical performance. No new or unexpected risks have been identified in the literature that would alter the established benefit–risk profile for short-term ureteral stents.

	BIORAD MEDISYS PVT. LTD.	Document No.	CER/DJS/61
		Revision. No	01
	CLINICAL EVALUATION REPORT PRODUCT- DOUBLE J STENT	Revision Date	26-12-2025

4 Device under evaluation

4.1 Route of evaluation

The conformity with GSPRs was demonstrated in accordance with Article 61(3) of the EU MDR 2017/745 regulation, by proving equivalence as outlined in Annex XIV.

The route of evaluation involved comparing the subject device to an equivalent device with the same technological, biological, and clinical characteristics. Clinical data from the equivalent device, such as clinical trials, published literature, or post-market data, was reviewed to establish equivalence in safety and performance. This data supported that the subject device performed similarly to the equivalent device within its intended use. The supporting data included preclinical data, which was used to provide evidence of the device's functionality, biocompatibility, and overall risk profile.

In this clinical evaluation, the Double Pigtail Ureteral Stent (Cook Medical) device has been regarded as the equivalent device (**Section 3.6**). Equivalence with the device in scope of this CER has been established by comparing technical, biological and clinical characteristics of the same with Black Silicone Filiform Double Pigtail Ureteral Stent device. Additionally, the supporting clinical data available for the Black Silicone Filiform Double Pigtail Ureteral Stent device has been utilized to establish acceptable thresholds for clinical safety and performance parameters.

Demonstration of equivalence

The Double J Stent (Short term) is based on established and proven technology currently available on the market. An equivalent, market-proven device has been selected for this comparison: Double Pigtail Ureteral Stent (Cook Medical)

4.2.1 Equivalency Comparison

Equivalency has been demonstrated across three key aspects:

- Technical Equivalency
- Biological Equivalency
- Clinical Equivalency

	BIORAD MEDISYS PVT. LTD.	Document No.	CER/DJS/61
		Revision. No	01
	CLINICAL EVALUATION REPORT PRODUCT- DOUBLE J STENT	Revision Date	26-12-2025

Table 14: Equivalence Table

Category	Characteristic	Device 1 Biorad – Double J Stent (Short Term)	Device 2 Cook Medical – Double Pigtail Ureteral Stent	Identified Differences / Conclusion	Scientific Justification / Impact on Safety & Performance	Clinical Significant Difference (Yes/No)
Technical	Design & Operating Principle	Passive ureteral stent with bilateral pigtail (Double J) configuration; internal urinary drainage	Passive double pigtail ureteral stent; internal urinary drainage	No fundamental difference in design principle	Both devices maintain ureteral patency via hollow lumen and pigtail retention. Same mechanical action and drainage function.	No
	Mechanical Action	Maintains ureteral patency and facilitates urine flow from kidney to bladder	Maintains ureteral patency and facilitates urine flow	No difference in core mechanical function	Both achieve intended effect through identical passive mechanical support mechanism.	No
	Dimensions /	Multiple diameters and lengths;	Multiple diameters and lengths;	Minor dimensional	Dimensional ranges fall within standard	No

	BIORAD MEDISYS PVT. LTD.	Document No.	CER/DJS/61
		Revision. No	01
	CLINICAL EVALUATION REPORT PRODUCT- DOUBLE J STENT	Revision Date	26-12-2025

	Configurati on	tapered ends; drainage holes	drainage holes	I variation possible	ureteral stent specifications and do not alter clinical function.	
Biologic al	Material in Contact with Tissue	Polyuretha ne (patient- contacting component)	Polymer- based ureteral stent material (e.g., polyuretha ne or silicone, depending on model)	Equivalent class of biocompati ble polymer	Both use established, ISO 10993- evaluated materials for short-term urinary tract contact. No novel substances introduced.	No*
	Type & Duration of Contact	Short-term indwelling ureteral contact (≤30 days)	Temporary indwelling ureteral contact	Equivalent contact type and duration category	Same anatomical site, same mucosal contact, same fluid environment (urine).	No
	Release Characterist ics	Non-drug eluting; no intended substance release	Non-drug eluting (standard model assumed)	No difference	Both are passive devices without pharmacolog ical action.	No
Clinical	Intended Purpose	Internal urinary drainage in ureteral obstruction and post-	Internal urinary drainage in ureteral obstructio n and	Same intended purpose	Same clinical objective: maintain ureteral patency and	No

	BIORAD MEDISYS PVT. LTD.	Document No.	CER/DJS/61
		Revision. No	01
	CLINICAL EVALUATION REPORT PRODUCT- DOUBLE J STENT	Revision Date	26-12-2025

		urological procedures	post-urological procedures		ensure drainage.	
	Anatomical Site	Ureter (renal pelvis to bladder)	Ureter (renal pelvis to bladder)	No difference	Same site of implantation and physiological environment.	No
	Target Population	Adult patients requiring ureteral drainage	Adult patients requiring ureteral drainage (depending on model)	Equivalent population	Similar anatomy, disease condition, and clinical setting.	No
	Intended User	Urologist / trained clinician	Urologist / trained clinician	No difference	Same professional user group and clinical setting.	No
	Expected Clinical Performance	Effective short-term drainage; reduction of obstruction ; pigtail retention reduces migration	Effective short-term drainage; pigtail retention reduces migration	No difference in expected outcome	Literature for ureteral stents supports comparable safety and performance for devices of similar design and material.	No

	BIORAD MEDISYS PVT. LTD.	Document No.	CER/DJS/61
		Revision. No	01
	CLINICAL EVALUATION REPORT PRODUCT- DOUBLE J STENT	Revision Date	26-12-2025

4.2 Justification for WET

The Double J Stent (Short Term) represents a well-established technology within the field of urology. Ureteral stents incorporating a bilateral pigtail (“Double J”) configuration have been in routine clinical use for several decades as a standard method for temporary internal urinary drainage. Since their introduction in the late 20th century, Double J ureteral stents have become an integral component of contemporary endourological practice for the management of ureteral obstruction, postoperative drainage following ureteroscopy or lithotripsy, and protection of ureteral integrity after reconstructive procedures. The fundamental design principle and clinical application of these devices have remained stable over time, and their safety and performance characteristics are extensively documented in scientific literature.

The systematic reviews and meta-analyses included in the literature review (Haleblian et al., 2008; Nabi et al., 2007; Song et al., 2012) confirm that ureteral stents are widely accepted as effective devices for maintaining ureteral patency and preventing postoperative obstruction. These publications consistently demonstrate that stents provide reliable internal drainage when clinically indicated and are associated with predictable and manageable adverse effects. The literature further establishes that ureteral stenting is a common adjunct to ureteroscopy and stone management, reflecting its integration into the standard of care. Survey data (Auge et al., 2007) and expert discussions (Hughes et al., 2014) reinforce that ureteral stent placement is a routine clinical decision guided by patient and procedural factors, further supporting the maturity and established nature of the technology.

The mechanism of action of the Double J Stent (Short Term) is purely mechanical and passive. The device maintains ureteral patency by providing a hollow conduit between the renal pelvis and bladder, thereby facilitating urine drainage and preventing obstruction due to ureteral edema, calculi fragments, or anatomical narrowing. The bilateral pigtail configuration at both ends of the stent serves as a retention mechanism to reduce migration and ensure positional stability. This mechanism of action has not changed over decades of use and is consistently described in the reviewed literature. No pharmacological, immunological, or metabolic mode of action is involved. The absence of an active component further supports the categorization of the device as a mature and well-characterized technology.

	BIORAD MEDISYS PVT. LTD.	Document No.	CER/DJS/61
		Revision. No	01
	CLINICAL EVALUATION REPORT PRODUCT- DOUBLE J STENT	Revision Date	26-12-2025

The safety profile of ureteral stents is similarly well established. The reviewed publications, including Joshi et al. (2003), demonstrate that irritative lower urinary tract symptoms, flank discomfort, and mild hematuria are common but generally transient and manageable during the indwelling period. Meta-analyses (Nabi et al., 2007; Song et al., 2012) report that while stent placement may increase postoperative urinary symptoms in some patient groups, serious adverse events remain uncommon. Additional case reports and clinical series included in the literature review describe complications such as encrustation, migration, obstruction, and fragmentation, particularly in the context of prolonged or forgotten stents. Importantly, these complications are strongly associated with extended indwelling duration beyond the intended short-term use rather than with inherent device design flaws. The predictable, time-dependent nature of these risks confirms that they are intrinsic to the device category and are adequately mitigated through appropriate patient selection, correct placement, and timely removal.

The Double J Stent (Short Term) does not introduce novel materials, coatings, drug-eluting properties, or design features that would alter the established mechanism of action or risk profile of conventional ureteral stents. The materials used are established biocompatible polymers commonly employed in ureteral stent manufacturing and evaluated in accordance with ISO 10993 standards. The device does not present new clinical uncertainties compared to other marketed double pigtail ureteral stents. Therefore, from a technical, biological, and clinical perspective, the subject device aligns with the well-characterized device category described extensively in the literature.

In accordance with Article 61(10) of Regulation (EU) 2017/745, a clinical investigation is not required where sufficient clinical evidence is available through existing literature, equivalence demonstration, post-market surveillance data, and risk management documentation. The systematic reviews, meta-analyses, observational studies, and case series included in the 12 selected articles collectively provide robust and consistent evidence regarding the safety, performance, and known side-effect profile of ureteral stents. The data demonstrate reproducible clinical outcomes and confirm that the benefit–risk profile of short-term ureteral stenting remains favorable when used as intended.

	BIORAD MEDISYS PVT. LTD.	Document No.	CER/DJS/61
		Revision. No	01
	CLINICAL EVALUATION REPORT PRODUCT- DOUBLE J STENT	Revision Date	26-12-2025

Considering the longstanding clinical history, stable and well-understood mechanism of action, predictable safety profile, absence of novel technological characteristics, and extensive supportive scientific literature, the Double J Stent (Short Term) fulfills the criteria of a Well-Established Technology (WET). The available clinical evidence is therefore sufficient to support conformity with the applicable General Safety and Performance Requirements (GSPRs) of Regulation (EU) 2017/745 without the need for a new pre-market clinical investigation. Ongoing post-market surveillance and periodic clinical evaluation updates ensure continued confirmation of safety and performance throughout the device lifecycle.

4.2.1 Equivalency Conclusion

Based on the comparative assessment of technical, biological, and clinical characteristics, equivalence between the Biorad Double J Stent (Short Term) and the Cook Medical Double Pigtail Ureteral Stent has been established in accordance with the requirements of MDR Annex XIV (Part A) and MDCG 2020-5 guidance.

Both devices share the same fundamental design concept (bilateral pigtail ureteral stent), operating principle (passive internal urinary drainage), and mechanical action (maintenance of ureteral patency). The materials used are established biocompatible polymers intended for short-term indwelling ureteral contact, with similar type and duration of tissue interaction and no intended pharmacological or biological mode of action. Clinically, both devices are intended for the same purpose—temporary internal drainage of the ureter in conditions such as obstruction or post-urolurgical procedures—used in the same anatomical site, patient population, and by the same type of healthcare professionals.

No clinically significant differences have been identified that would impact the safety or clinical performance of the subject device when compared to the selected marketed device. Therefore, it is concluded that the Biorad Double J Stent (Short Term) is equivalent to the Cook Medical Double Pigtail Ureteral Stent for the intended purpose, and the clinical data available for the equivalent device are applicable to support the clinical evaluation of the subject device.

	BIORAD MEDISYS PVT. LTD.	Document No.	CER/DJS/61
		Revision. No	01
	CLINICAL EVALUATION REPORT PRODUCT- DOUBLE J STENT	Revision Date	26-12-2025

4.2.2 Conformity Assessment Summary

The Double J Stent (Short Term) is classified as a Class IIa medical device in accordance with the classification rules set out in Annex VIII of Regulation (EU) 2017/745 (EU MDR), as it is an invasive device introduced into the body via a natural orifice and intended for short-term use. The conformity assessment procedure for the device has been carried out in accordance with Annex IX of the EU MDR (Conformity Assessment Based on a Quality Management System and Assessment of Technical Documentation). The manufacturer has established and maintains a Quality Management System compliant with Article 10(9) of the MDR and EN ISO 13485, covering design, manufacturing, final inspection, and post-market activities.

The Technical Documentation has been prepared in accordance with Annex II and Annex III of the MDR and includes the device description and specifications, design and manufacturing information, risk management documentation, verification and validation data (including biological evaluation, sterilization validation, packaging validation, and performance testing), labeling and IFU, as well as the Clinical Evaluation Report prepared in accordance with Annex XIV (Part A).

For Class IIa devices, the Notified Body has assessed the manufacturer's Quality Management System and reviewed the relevant Technical Documentation as required under Annex IX. Following successful assessment, the device bears the CE marking in accordance with Article 20 of the MDR.

Ongoing conformity is ensured through implementation of Post-Market Surveillance (PMS) and vigilance activities in accordance with Articles 83–86 and Annex III, including periodic review of the clinical benefit–risk profile and updating of the Clinical Evaluation as necessary.

4.3 Data generated and held by the manufacturer

The following data held by Biorad were collected and analysed for the Double J Stent (Short term):

- Pre-clinical data
 - Biocompatibility tests
 - Bench tests
- Clinical data

	BIORAD MEDISYS PVT. LTD.	Document No.	CER/DJS/61
		Revision. No	01
	CLINICAL EVALUATION REPORT PRODUCT- DOUBLE J STENT	Revision Date	26-12-2025

- Adverse events data from several databases for similar/generic devices
- Data from literature
- PMCF data

4.3.1 Pre-clinical data

Pre-clinical data generated and held by the manufacturer includes biocompatibility and bench testing data.

	BIORAD MEDISYS PVT. LTD.	Document No.	CER/DJS/61
		Revision. No	01
	CLINICAL EVALUATION REPORT PRODUCT- DOUBLE J STENT	Revision Date	26-12-2025

4.3.1.1 Biocompatibility data

Biocompatibility tests have been conducted for the Double J Stent/Kit (Long Term) to ensure the safety of the material. The device has passed the biocompatibility tests following the requirements of ISO 10993-1:2018. The tests cited below were conducted on the final finished device.

The device components that come into direct contact with human tissue are tested for biocompatibility as per ISO 10993 standards, ensuring

Table 14: Details of Biocompatibility tests

Test	Applicable Standard	Objective	Report Number	Results/Conclusion
In-Vitro Cytotoxicity	ISO 10993-5	To evaluate the cytotoxic potential of test item when treated using L929 cell line.	ARL/G/22/PT0287	Non-cytotoxic in both undiluted (100%) and all diluted concentrations (50%, 25%, 12.5%, 6.25%) under the specified test conditions.
Skin Sensitization	ISO 10993-10	To evaluate the potential for skin sensitization	ARL/G/22/PT0288	Did not induce skin sensitization; considered a non-sensitizer under study conditions.
Acute Systemic Toxicity	ISO 10993-11	To evaluate acute systemic toxicity potential of polar and non-	ARL/G/22/PT0291	No systemic toxicity observed with polar and non-polar extracts.

	BIORAD MEDISYS PVT. LTD.	Document No.	CER/DJS/61
		Revision. No	01
	CLINICAL EVALUATION REPORT PRODUCT- DOUBLE J STENT	Revision Date	26-12-2025

		polar extracts of the test item, when administered as a single dose through Intravenous and Intraperitoneal route to swiss albino mice.		
Chemical Characterization	ISO 10993-18 & ISO 10993-17	To perform chemical characterization study and estimation of allowable limit of leachables for test item.	ARL/G/22/AD0286	No significant levels of different leachates observed in the present study.
Intracutaneous Reactivity	ISO 10993-23	To evaluate possible local tissue reaction with extracts when injected by intracutaneous route in white rabbits.	ARL/G/22/PT0289	No irritation observed; test item found to be non-irritant.
Pyrogenicity	ISO 10993-11	To evaluate the changes in rectal temperature following single dose intravenous injection of the	ARL/G/22/PT0290	Non-pyrogenic under the test conditions.

	BIORAD MEDISYS PVT. LTD.	Document No.	CER/DJS/61
		Revision. No	01
	CLINICAL EVALUATION REPORT PRODUCT- DOUBLE J STENT	Revision Date	26-12-2025

		test item to New Zealand white rabbits.		
In vitro Mammalian Cell Micronucleus	ISO 10993-3	To evaluate the clastogenic and aneugenic potential of test item.	ARL/G/22/PT0293	Non-clastogenic and non-aneugenic at undiluted of test item extract (100%) under test conditions.
Bacterial Reverse Mutation	ISO 10993-3	To evaluate genotoxic potential of the test item.	ARL/G/22/PT0292	Non-mutagenic under the test conditions.
Sub-acute and Sub-chronic Toxicity	ISO 10993-11, ISO 10993-6	To evaluate possible local tissue reaction and health hazards that arise from a prolonged exposure of the test item.	ARL/G/22/PT0294	No systemic toxicity or adverse reaction under test conditions.

Based on the results of the conducted biocompatibility evaluations, the Double J Stent/Kit (Long Term)–Double J Stent/Kit (Long Term) (Disposable Insertion Tube) demonstrated compliance with all relevant ISO 10993 standards. The device was found to be non-cytotoxic, non-sensitizing, non-irritant, non-pyrogenic, and non-toxic systemically under the tested conditions. Furthermore, chemical characterization and toxicological risk assessments confirmed that all leachable elements and volatile organic compounds were within acceptable limits, indicating the material's chemical safety for clinical use. Overall, the test item was confirmed to be biocompatible and safe for its intended use as a medical device in humans.

	BIORAD MEDISYS PVT. LTD.	Document No.	CER/DJS/61
		Revision. No	01
	CLINICAL EVALUATION REPORT PRODUCT- DOUBLE J STENT	Revision Date	26-12-2025

4.3.1.2 Bench testing (performance testing)

Performance tests have been conducted for the Double J Stent/Kit (Long Term) to ensure the device performs as intended and meets acceptance criteria according to regulatory standards and in-house limits. All the tests are performed based on an internal Standard Operating Procedure (SOP) in reference to international standards.

Biorad has performed bench marking study for the Double J Stent/Kit (Long Term) during product realization phase and compared it with several other CE marked Double J Stent/Kit (Long Term) available in the market.

Table 19: Bench marking study

Sr. No.	Product Details		Manufacturers	
			Biorad Medisys	Cook Medical
			1	2
1	Batch Number		BA01016107	C1261537
2	Manufacturing Date		19-10-16	10-08-2016
3	Expiry Date		19-09-19	10-08-2019
	Parameters	Limits (For Biorad Products)		
1	Color	NA	White	White
2	Outer Diameter	1.95-2.05	1.95mm	1.98mm
3	Length	±0.5cm	26.2	26.1
4	ID	1.15-1.19	1.17	1.20
5	Kinking of Product	No Kinking	No Kinking	No Kinking
6	Tensile Strength	≥ 5N	22.5N,197.98mm Elongated Length	22N, 256.86mm Elongated Length
7	Packaging	The maintenance of sterility of the contents under dry, clean and adequately ventilated storage conditions	The products are sealed in Primary Packing which is followed by secondary packing done with Tyvek pouch which is highly puncture resistant	The products are packed in blister which is followed by secondary packing done with Tyvek pouch
		Adequate protection of the contents during normal handling, transit and storage.		
		The minimum risk of contamination of the contents during removal from the container		
8	Labelling Information	Size of the Product	Present	Present
		Symbols as per	BS EN ISO 15223-1 2012	-

Based on the above-mentioned study, it can be inferred that the subject device is equivalent to Black Silicone Filiform Double Pigtail Ureteral Stent. (Cook Medical)

	BIORAD MEDISYS PVT. LTD.	Document No.	CER/DJS/61
		Revision. No	01
	CLINICAL EVALUATION REPORT PRODUCT- DOUBLE J STENT	Revision Date	26-12-2025

4.3.2 Clinical data

4.3.2.1 Clinical investigations

The Double J Stent/Kit (Short Term) is an implantable as it is inserted into the body for several days or weeks and then removed, this technology has indeed been in routine clinical use for over a decade. Hence, according to MDR article 61, no clinical investigation was performed.

4.3.2.2 Post-market surveillance data

The device is marketed in India. The following data sources were utilised to collect relevant post-marketing data to support clinical safety and performance.

- Adverse event data of generic devices
- Post-market clinical follow-up data of device under evaluation

A. Distribution details

Under Biorad Medisys Private Limited in India the company received its manufacturing license approved by CDSCO (India). A total of 1859 units of the device were sold in India between Jan 2025, and Dec 2025 & 1819 units of the device were sold in EU Countries between Jan 2025 and Dec 2025. Below is the summary of the distribution data:

Table 15: Distribution pattern

	BIORAD MEDISYS PVT. LTD.	Document No.	CER/DJS/61
		Revision. No	01
	CLINICAL EVALUATION REPORT PRODUCT- DOUBLE J STENT	Revision Date	26-12-2025

Year	Months wise distribution	Units Sold	Sales in EU Countries (If applicable)	Sales in Non-EU Countries (If applicable)
2025	Jan	78	45	93
	Feb	102	64	143
	Mar	107	23	127
	Apr	96	87	89
	May	146	165	155
	June	189	55	197
	July	176	223	171
	Aug	195	421	198
	Sep	202	187	225
	Oct	138	43	140
	Nov	124	472	136
	Dec	160	34	185

B. Complaints & Feedback

The Manufacturer has provided the subject device to 5 healthcare professionals for stenting application and asked for their feedback. All of them given 'Good' rating against 'product satisfaction'. An overall rating of 'Very Good' was given by 4 of them and the other clinician rated as 'Excellent'. A total number of 250 patients were treated during this process. However, 8 incidents of complications related to the application of the device was reported by those professionals: 4 patients (7.14 %) had Ureteral Erosion /Perforation, 3 patients (6.12 %) had Urinary Tract Infection (UTI), 5 patients (8.33 %) had Stent Malposition, and 2 patient (4.34 %) had Ureteral Stent Obstruction.

Table 16: Feedback details

Sr. No.	Doctor	Hospital	City	Count of Patients	Post Procedural Side Effects (in Patients)
1.	Dr. Prashant Pattnaik	Sahyadri Super Speciality	Mumbai	56	4 (Ureteral Erosion /Perforation)

	BIORAD MEDISYS PVT. LTD.	Document No.	CER/DJS/61
		Revision. No	01
	CLINICAL EVALUATION REPORT PRODUCT- DOUBLE J STENT	Revision Date	26-12-2025

		Hospital Hadapsar			
2.	Dr. Suraj Lunavat	Sahyadri Hospital	Nashik	49	3 (Urinary Tract Infection)
3.	Dr. Amol Taulikar	Aditya Birla Memorial Hospital	Jalgaon	60	5 (Stent Malposition)
4.	Dr. Hiralal Chaudhari	Urology Clinic	Nagpur	39	---
5.	Dr. Sayed Mohammed	Sahyadri Super Speciality Hospital Hadapsar	Amravati	46	2 (Ureteral Stent Obstruction)

Based on the feedback, the user was satisfied with the use of the Double J Stent/Kit (Long Term). The complications can be caused due to the entire procedure and not only due to the Double J Stent/Kit (Long Term).

No complaints have been received to date for the products sold in the Indian (non-EU) market.

C. Adverse event data of similar devices from adverse event databases

Adverse event databases were accessed as planned. The data were collected for the period January 2005 to June 2025. The results obtained from various databases are summarised below:

- **Summary of results from Recall database, United States Food and Drug Administration (US FDA)**

A search was performed on the USFDA Recalls Database for the product code (FPA). Data for medical device recalls and search results are described in Error! Reference source not found..

Table 17: USFDA Recalls data search

Date of Search: 19/06/2025

Review Period: 01/01/2005 to 19/06/2025

	BIORAD MEDISYS PVT. LTD.	Document No.	CER/DJS/61
		Revision. No	01
	CLINICAL EVALUATION REPORT PRODUCT- DOUBLE J STENT	Revision Date	26-12-2025

Product Code	Number of results obtained	Link for the database
FAD	27	USFDA Recalls Database

Table 18: Included Results of Product Recalls

Trade Name	Date	Manufacturer	Reason for Recall
Cook Inc C-Flex Double Pigtail Ureteral Stent Set	02/05/2021	COOK INC	Manufactured with incorrect material, which may result in the stent being stiffer and potentially susceptible to degradation could potentially lead to breakage/fragmentation.
<u>Contour VL</u> Ureteral Stent Set	12/30/2020	Boston Scientific Corporation	Trace amounts of metallic debris were found on the stent related to the manufacturing process. The debris was confirmed to be particles of nickel.
<u>Contour VL</u> Ureteral Stent Set	12/30/2020	Boston Scientific Corporation	Trace amounts of metallic debris were found on the stent related to the manufacturing process. The debris was confirmed to be particles of nickel.
Cook Medical Universa Soft Ureteral Stent Sets	09/15/2020	Cook Inc.	Universa Soft Ureteral Stent Sets were incorrectly placed in the outer boxes labeled as Universa Firm Ureteral Stent Set; pouch label reflects the packaged product.
Contour VL Ureteral Stent Set	04/01/2020	Boston Scientific	An increase in the rate of complaints for difficulty or inability to track over the guidewire, may result in a procedural delay due to the need to exchange the affected device
Percuflex Plus Ureteral Stent Set	04/01/2020	Boston Scientific	An increase in the rate of complaints for difficulty or inability to track over the guidewire, may result in a procedural delay due to the need to exchange the affected device

	BIORAD MEDISYS PVT. LTD.	Document No.	CER/DJS/61
		Revision. No	01
	CLINICAL EVALUATION REPORT PRODUCT- DOUBLE J STENT	Revision Date	26-12-2025

Percuflex Plus Ureteral Stent Set	04/01/2020	Boston Scientific	An increase in the rate of complaints for difficulty or inability to track over the guidewire, may result in a procedural delay due to the need to exchange the affected device
Stretch VL Ureteral Stent Set	04/01/2020	Boston Scientific	An increase in the rate of complaints for difficulty or inability to track over the guidewire, may result in a procedural delay due to the need to exchange the affected device
Stretch VL Ureteral Stent Set	04/01/2020	Boston Scientific	An increase in the rate of complaints for difficulty or inability to track over the guidewire, may result in a procedural delay due to the need to exchange the affected device
Stretch VL Ureteral Stent Set	04/01/2020	Boston Scientific	An increase in the rate of complaints for difficulty or inability to track over the guidewire, may result in a procedural delay due to the need to exchange the affected device

Note: Refer to Appendix-V for the Adverse Events Search Results.

- **Total Product Life Cycle (TPLC)¹ - USFDA**

TPLC approach allows the FDA to review and monitor medical devices throughout their life cycle by considering all available information on safety and effectiveness. data was available since 2010 for the product code of the Double J Stent/Kit (Long Term).

Table 19: Technical complications reported in TPLC

Device Problems	MDRs with this Device Problem
Break	759
Microbial Contamination of Device	302

	BIORAD MEDISYS PVT. LTD.	Document No.	CER/DJS/61
		Revision. No	01
	CLINICAL EVALUATION REPORT PRODUCT- DOUBLE J STENT	Revision Date	26-12-2025

<u>Adverse Event Without Identified Device or Use Problem</u>	249
<u>Failure to Clean Adequately</u>	211
<u>Device Reprocessing Problem</u>	164
<u>Mechanical Problem</u>	121
<u>No Display/Image</u>	106
<u>Detachment of Device or Device Component</u>	93
<u>Material Protrusion/Extrusion</u>	90
<u>Poor Quality Image</u>	87
<u>Material Split, Cut or Torn</u>	68
<u>Peeled/Delaminated</u>	65
<u>Fluid/Blood Leak</u>	48
<u>Contamination</u>	47
<u>Material Deformation</u>	46
<u>Leak/Splash</u>	46
<u>Material Puncture/Hole</u>	45
<u>Physical Resistance/Sticking</u>	42
<u>Mechanical Jam</u>	40
<u>Optical Problem</u>	39
<u>Material Twisted/Bent</u>	37
<u>Appropriate Term/Code Not Available</u>	32
<u>Material Separation</u>	30
<u>Contamination /Decontamination Problem</u>	29
<u>Scratched Material</u>	27
<u>Crack</u>	27
<u>Insufficient Information</u>	26
<u>Corroded</u>	26
<u>Difficult to Remove</u>	25

	BIORAD MEDISYS PVT. LTD.	Document No.	CER/DJS/61
		Revision. No	01
	CLINICAL EVALUATION REPORT PRODUCT- DOUBLE J STENT	Revision Date	26-12-2025

<u>Device Contamination with Chemical or Other Material</u>	24
<u>Display or Visual Feedback Problem</u>	22
<u>Material Fragmentation</u>	20
<u>Defective Component</u>	18
<u>Device Displays Incorrect Message</u>	18
<u>Use of Device Problem</u>	16
<u>Erratic or Intermittent Display</u>	16
<u>Unsealed Device Packaging</u>	16
<u>Loose or Intermittent Connection</u>	16
<u>Material Perforation</u>	15
<u>Fracture</u>	15
<u>Defective Device</u>	15
<u>Degraded</u>	15
<u>Component Missing</u>	12
<u>Material Integrity Problem</u>	12
<u>Device Damaged by Another Device</u>	11
<u>Melted</u>	11
<u>Device Handling Problem</u>	11
<u>Unintended Movement</u>	10
<u>Improper or Incorrect Procedure or Method</u>	10
<u>Device Operates Differently Than Expected</u>	9

The most common issues are break, microbial contamination of device, adverse event without identified device or use problem, failure to clean adequately, device reprocessing problem, mechanical problem, and no display/image. Other issues reported are detachment of device or device component, material protrusion/extrusion, poor quality image, material split, cut or torn, peeled/delaminated, fluid/blood leak, contamination, material deformation, leak/splash, material puncture/hole, physical resistance/sticking, mechanical jam, optical problem, material twisted/bent, appropriate term/code not available, material separation, scratched material, crack, insufficient information, corroded, difficult to remove.

	BIORAD MEDISYS PVT. LTD.	Document No.	CER/DJS/61
		Revision. No	01
	CLINICAL EVALUATION REPORT PRODUCT- DOUBLE J STENT	Revision Date	26-12-2025

- **Summary of results from Medicines and Healthcare Products Regulatory Agency (MHRA), United Kingdom (UK)**

A search was conducted on the MHRA database for the period of 01/01/2005 to 19/06/2025 using the keywords mentioned below. A total of 27 Field Safety Notices (FSN) were found. However, none were found relevant.

Table 20: Search results from MHRA

Date of Search: 19/06/2025				
Review Period: 01/01/2005 to 19/06/2025				
Search Keyword	Number of results obtained	Number of results included	Reason for exclusion	Link for the database
Double J Stent/Kit (Long Term)	00	00	N/A	MHRA search

- **Summary of results from Therapeutic Goods Administration (TGA), Australia**

A search was conducted on the TGA database using the keywords mentioned below. No results were obtained for any of the keywords.

Table 21: Search results from TGA

Date of Search: 19/06/2025				
Review Period: 01/01/2005 to 19/06/2025				
Search Keyword	Number of results obtained	Number of results included	Reason for exclusion	Link for the database
Double J Stent/Kit (Long Term)	0	N/A	N/A	TGA Search

	BIORAD MEDISYS PVT. LTD.	Document No.	CER/DJS/61
		Revision. No	01
	CLINICAL EVALUATION REPORT PRODUCT- DOUBLE J STENT	Revision Date	26-12-2025

Black Silicone Filiform Double Pigtail Ureteral Stent	0	N/A	N/A	
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- **Summary of results from Drug & Health Product Register (DHPR), Health Canada**

A search was conducted on the HPR database using the keywords mentioned below.

Table 22: Search results from DHPR

Date of Search: 19/06/2025				
Review Period: 01/01/2005 to 19/06/2025				
Search Keyword	Number of results obtained	Number of results included	Reason for exclusion	Link for the database
Double J Stent/Kit (Long Term)	00	00	N/A	DHPR Search

The recall was initiated due to a labelling update required to correct the instructions for use of certain Stent. The correction advises that high-level disinfection and, for specific models, liquid chemical sterilization should not be used as reprocessing methods. These methods were previously listed in error. Users without access to the appropriate sterilization method specified in the updated instructions are instructed to discontinue use and return the affected stent to the manufacturer (Cook Medical).

- **Summary of results from the Federal Institute for Drugs and Medical Devices (Bundesinstitut für Arzneimittel und Medizinprodukte, BfArM , Germany)**

A search was conducted on the BfArM database using the keywords mentioned below. The result obtained was not found relevant.

Table 23: Search results from BfArM

	BIORAD MEDISYS PVT. LTD.	Document No.	CER/DJS/61
		Revision. No	01
	CLINICAL EVALUATION REPORT PRODUCT- DOUBLE J STENT	Revision Date	26-12-2025

Date of Search: 19/06/2025				
Review Period: 01/01/2005 to 19/06/2025				
Search Keyword	Number of results obtained	Number of results included	Reason for exclusion	Link for the database
Double J Stent/Kit (Long Term)	00	00	Not related to the device in scope	BfArm

The FSNs (Field Safety Notice) were related to irrigation plug of Cook Medical.

- **Summary of results from Central Drugs Standard Control Organization (CDSCO), India**

A search was conducted on the CDSCO database using the keywords mentioned below. No results were obtained for any of the keywords.

Table 24: Search results from CDSCO

Date of Search: 19/06/2025				
Review Period: Not Applicable				
Search Keyword	Number of results obtained	Number of results included	Reason for exclusion	Link for the database
Double J Stent/Kit (Long Term)	00	N/A	N/A	CDSCO

Refer to Appendix V for Adverse Events Search Results.

D. Post-Market Clinical Follow-up (PMCF) data

Post-market clinical follow-up will be conducted for the subject device once the device is marketed in the EU region. Refer PMCF Plan-BRP/PMCFP/DJS/61 & PMCF Report-BRP/PMCF/DJS/61

	BIORAD MEDISYS PVT. LTD.	Document No.	CER/DJS/61
		Revision. No	01
	CLINICAL EVALUATION REPORT PRODUCT- DOUBLE J STENT	Revision Date	26-12-2025

Literatures obtained from initial search and systematic search were utilised to identify pertinent information available for the Double J Stent/Kit (Long Term), to determine whether the device under evaluation is state-of-the-art, and to evaluate clinical safety and performance. The results from all the literature databases are given in the table below.

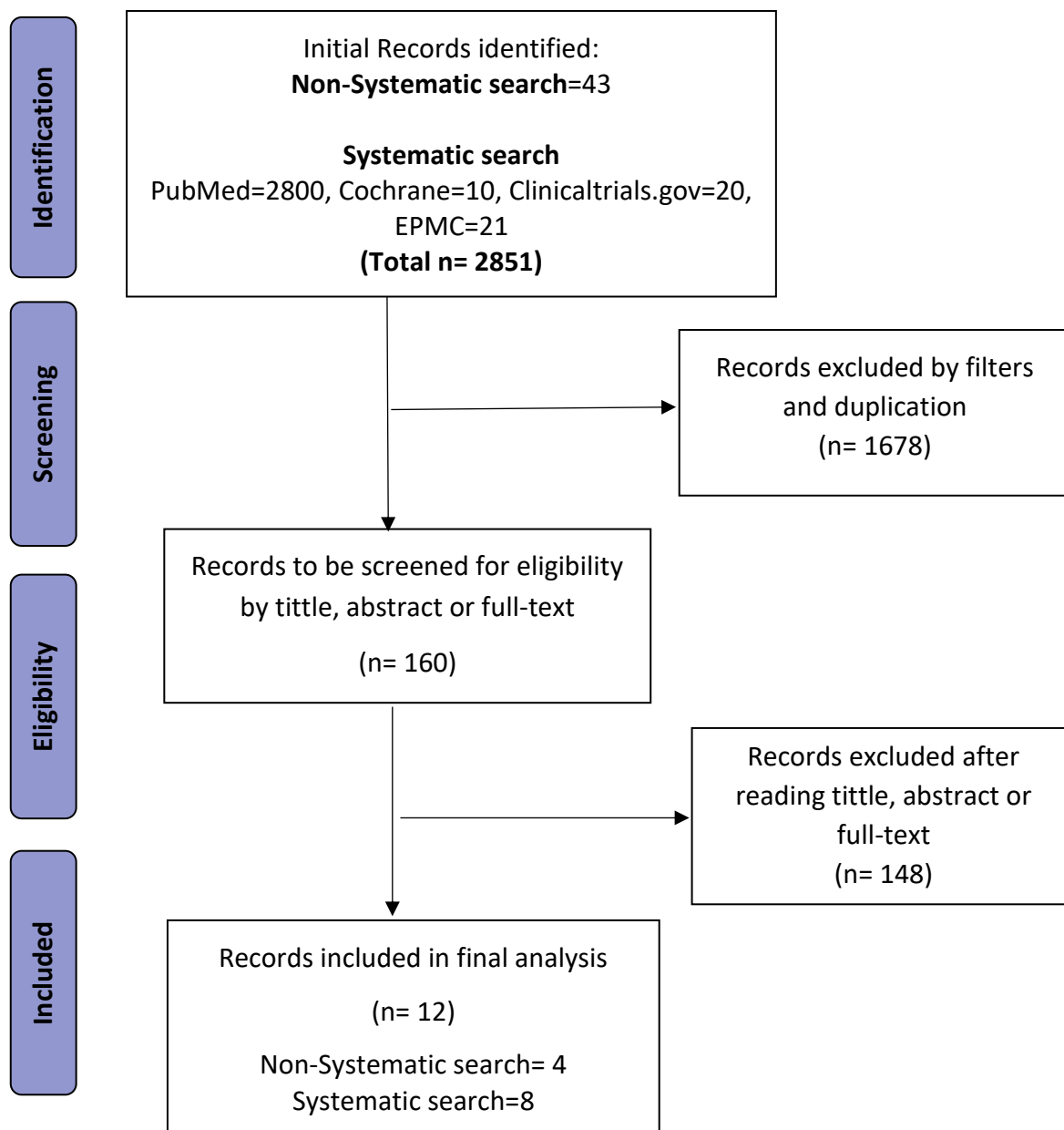
Table 25: Cumulative Literature Search Data from all sources

Database	Hits After Filters & Removal of Duplicates	Excluded	Included
Non-systematic search			
Google Scholar	43	39	4
Systematic Search			
PubMed	88	86	02
Cochrane Reviews	03	03	00
Clinicaltrials.Gov	05	03	02
EPMC	21	17	04
Total	160	148	12

Refer to Appendix I - Literature Search Protocol and Appendix II - Literature Search Report for further details.

	BIORAD MEDISYS PVT. LTD.	Document No.	CER/DJS/61
		Revision. No	01
	CLINICAL EVALUATION REPORT PRODUCT- DOUBLE J STENT	Revision Date	26-12-2025

PRISMA Flow chart: Preferred Reporting Items for Systematic Reviews and Meta-Analyses



4.4 Critical appraisal of clinical data

4.5.1 Appraisal method and criteria

	BIORAD MEDISYS PVT. LTD.	Document No.	CER/DJS/61
		Revision. No	01
	CLINICAL EVALUATION REPORT PRODUCT- DOUBLE J STENT	Revision Date	26-12-2025

Table 26: Appraisal criteria

Appraisal criteria		Description of criteria	Appraisal score
Appraisal criteria for suitability			
Appropriate Device	D1	Data reported from subject device	3
	D2	Data reported from equivalent device(s)	2
	D3	Data reported from similar devices, current knowledge/state-of-the-art, medical alternatives	1
Appropriate Device Application	A1	Device applications as intended by the manufacturer	3
	A2	Device use other than the intended applications by the manufacturer	2
	A3	No application mentioned	1
Appropriate Patient Group	P1	All patient groups intended by the manufacturer	3
	P2	Limited patient group	2
	P3	Different patient group; No patient group reported	1
Acceptable Report/Data Collation	R1	High quality reports or collations of data that contains sufficient information to undertake a rational and objective assessment	3
	R2	Reports with minor deficiencies	2
	R3	Reports with insufficient information to conclude on safety/performance	1
Appraisal criteria for data contribution			
Data Source Type	T1	Randomised controlled trials, meta-analyses, systematic reviews	3
	T2	Non-randomised studies, retrospective and prospective cohort studies, observational studies, registries	2
	T3	Case reports, Narrative reviews	1
Outcome Measures	O1	Reflect intended performance of the device	3
	O2	Do not reflect intended performance of the device or performance measures not reported	2
	O3	Unable to determine or performance measures not reported	1
Follow-up	F1	Duration of follow-up is appropriate to identify treatment effects and/or complications	3

	BIORAD MEDISYS PVT. LTD.	Document No.	CER/DJS/61
		Revision. No	01
	CLINICAL EVALUATION REPORT PRODUCT- DOUBLE J STENT	Revision Date	26-12-2025

Appraisal criteria		Description of criteria	Appraisal score
	F2	Duration of follow-up is not appropriate to identify treatment effects and/or complications	2
	F3	Duration of follow-up not reported or unable to determine	1
Statistical Significance	S1	Appropriate statistical analysis, statistical analysis not required due to the presentation of qualitative data (e.g., a case study)	3
	S2	Statistical analysis not appropriate	2
	S3	No statistics reported or unable to determine	1
Clinical Significance	C1	Magnitude of the treatment effect is clinically significant	3
	C2	Magnitude of the treatment effect is not clinically significant	2
	C3	No clinical effect is reported or unable to determine	1

Table 27: Appraisal scores

Level	Total score	Description
1	1	Sufficient clinical data for the subject device
2	2	Clinical data with gaps on subject device, equivalent and/or similar devices
3	3	Clinical data available for generic devices; information from guidance documents and narrative review articles available for state-of-the-art and alternative treatments

	BIORAD MEDISYS PVT. LTD.	Document No.	CER/DJS/61
		Revision. No	01
	CLINICAL EVALUATION REPORT PRODUCT- DOUBLE J STENT	Revision Date	26-12-2025

4.5.2 Appraisal of selected articles

The table below summarises appraisal criteria applied to all literatures included for evaluation of Double J Stent (Short term):

Appraisal Criteria for Suitability and Data Contribution

Indwelling ureteral stents: evaluation of symptoms, quality of life and utility (Joshi HB et al., 2003)

Parameter – Score – Justification

Parameter	Score	Justification
D1	3	Data from ureteral stents (device category directly relevant)
A1	3	Addresses intended use context (indwelling ureteral stent symptoms/QoL)
P1	3	Relevant patient population for ureteral stenting
R1	3	Provides clinically relevant safety/tolerability outcomes
T2	2	Observational clinical study (not RCT)
O1	3	Outcomes reflect clinical performance/tolerability (symptoms, QoL)
F1	3	Follow-up appropriate for symptom assessment in indwelling stent use
S1	3	Appropriate analysis for endpoints reported
C1	3	Clinically meaningful conclusions on symptom burden and QoL impact
Total	26	—
Level	2	—

Literature Summary:

This clinical study evaluates the symptom profile and quality-of-life (QoL) impact associated with indwelling ureteral stents. The authors systematically assess stent-related morbidity and demonstrate that ureteral stents, while clinically effective for internal drainage, are frequently associated with patient-reported symptoms that can affect daily functioning and wellbeing. The study highlights that common stent-associated effects include irritative lower urinary tract symptoms (e.g., frequency/urgency/dysuria), pain/discomfort, and reduced functional

	BIORAD MEDISYS PVT. LTD.	Document No.	CER/DJS/61
		Revision. No	01
	CLINICAL EVALUATION REPORT PRODUCT- DOUBLE J STENT	Revision Date	26-12-2025

capacity. Importantly, the publication provides evidence that stent tolerability is a meaningful clinical consideration and supports the need for patient counselling regarding expected transient symptoms during short-term indwelling use. The study contributes to the clinical background for ureteral stents by establishing that symptom burden and QoL changes are predictable, measurable outcomes for this device category and should be weighed in benefit–risk considerations.

Meta-analysis of postoperatively stenting or not in patients underwent ureteroscopic lithotripsy (Song T et al., 2012)

Parameter	Score	Justification
D1	3	Ureteral stenting directly evaluated
A1	3	Addresses peri-procedural use of ureteral stents (clinical context of use)
P1	3	Applicable patient population undergoing ureteroscopic lithotripsy
R1	3	Reports safety and symptomatic outcomes (benefits/risks)
T3	1	Meta-analysis (secondary evidence; dependent on included trial quality)
O1	3	Outcomes relevant to clinical performance and adverse effects
F1	3	Follow-up adequate for post-procedure outcomes
S1	3	Statistical synthesis performed
C1	3	Provides clinically significant conclusions on stent vs no-stent
Total	25	—
Level	1–2	Meta-analysis; evidence level depends on included study quality

Literature Summary:

This meta-analysis compares outcomes in patients who received a ureteral stent versus those managed without routine postoperative stenting following ureteroscopic lithotripsy. The analysis pools data from comparative clinical studies and evaluates postoperative symptoms and complications relevant to ureteral stent use, such as pain, urinary symptoms, infection, and the need for unplanned interventions. The results support that stenting is not universally

	BIORAD MEDISYS PVT. LTD.	Document No.	CER/DJS/61
		Revision. No	01
	CLINICAL EVALUATION REPORT PRODUCT- DOUBLE J STENT	Revision Date	26-12-2025

required in uncomplicated cases and that clinical decision-making should be indication-driven, considering procedural factors (e.g., ureteral trauma, residual fragments, edema, or risk of obstruction). The publication is valuable for state-of-the-art analysis because it demonstrates how ureteral stents are applied in practice as a protective and drainage-supporting measure, while also acknowledging that stent placement can introduce predictable symptom burden. The meta-analysis supports a balanced view in clinical evaluation: stents provide clear drainage benefits in indicated patients, but their use should be individualized to minimize unnecessary morbidity.

Ureteral stenting and urinary stone management: a systematic review (Haleblian G et al., 2008)

Parameter	Score	Justification
D1	3	Directly about ureteral stenting
A1	3	Addresses intended clinical role in stone management
P1	3	Relevant population undergoing stone-related urology care
R1	3	Summarizes expected benefits and known complications
T3	1	Systematic review (secondary evidence)
O1	3	Outcomes relevant to safety/performance and clinical practice
F1	3	Appropriate for summarizing clinical outcomes
S1	3	Review methodology and synthesis
C1	3	Clinically meaningful conclusions on indications/complications
Total	25	—
Level	1–2	Systematic review (depends on included evidence quality)

Literature Summary:

This systematic review examines the role of ureteral stents in urinary stone management, including pre-procedural, intra-procedural, and post-procedural use. The review consolidates clinical practice evidence on when stenting is beneficial (e.g., maintaining drainage in obstruction, facilitating ureteral healing after manipulation, preventing obstruction due to edema, and supporting postoperative recovery). It also discusses known limitations and

	BIORAD MEDISYS PVT. LTD.	Document No.	CER/DJS/61
		Revision. No	01
	CLINICAL EVALUATION REPORT PRODUCT- DOUBLE J STENT	Revision Date	26-12-2025

complications associated with stent use, including urinary symptoms, pain/discomfort, hematuria, infection, encrustation (particularly with prolonged dwelling), and occasional migration. By synthesizing multiple studies, the review establishes a comprehensive state-of-the-art baseline for ureteral stents: they are widely used and effective for internal urinary drainage, and their safety profile is well characterized with predictable and manageable side effects. This source supports CER sections describing current clinical practice, expected outcomes, and known complication patterns, reinforcing that the benefit–risk profile of short-term ureteral stenting is acceptable when used appropriately.

Outcomes of stenting after uncomplicated ureteroscopy: systematic review and meta-analysis (Nabi G et al., 2007)

Parameter	Score	Justification
D1	3	Directly evaluates ureteral stenting after ureteroscopy
A1	3	Matches common use case (post-ureteroscopy short-term stenting)
P1	3	Relevant patient population in ureteroscopy
R1	3	Reports complications and symptom outcomes
T3	1	Systematic review/meta-analysis
O1	3	Relevant to performance and adverse events
F1	3	Follow-up adequate for post-procedure events
S1	3	Statistical synthesis performed
C1	3	Clinically significant conclusions on routine vs selective stenting
Total	25	—
Level	1–2	Meta-analysis; depends on included trials

Literature Summary:

This systematic review and meta-analysis evaluates whether routine ureteral stenting is necessary after uncomplicated ureteroscopy and compares clinical outcomes between stented and non-stented patient groups. The publication focuses on clinically important endpoints such as postoperative pain, urinary symptoms, infection, unplanned readmissions, and need for secondary interventions. The pooled findings support that routine stenting after

	BIORAD MEDISYS PVT. LTD.	Document No.	CER/DJS/61
		Revision. No	01
	CLINICAL EVALUATION REPORT PRODUCT- DOUBLE J STENT	Revision Date	26-12-2025

uncomplicated ureteroscopy may not provide meaningful clinical benefit for all patients and may increase symptom burden, reinforcing selective use based on clinical judgement and patient risk factors. Importantly, the paper characterizes the typical postoperative issues associated with stents as part of the known risk profile of the device category, rather than unexpected safety concerns. For CER purposes, this meta-analysis provides strong state-of-the-art evidence supporting (i) established clinical utility of ureteral stents in indicated situations and (ii) the accepted trade-off between drainage benefits and manageable stent-related side effects, supporting a benefit–risk discussion consistent with MDR expectations.

Practice patterns of ureteral stenting after routine ureteroscopic stone surgery: survey (Auge BK et al., 2007)

Parameter	Score	Justification
D2	2	Survey on ureteral stent use (indirect clinical evidence)
A1	3	Relevant to real-world use patterns of ureteral stents
P1	3	Addresses clinician practice across typical patient settings
R2	2	Provides context but limited direct safety/performance outcomes
T3	1	Survey study (non-interventional)
O2	2	Practice-pattern outcome (not patient outcome)
F2	2	Not focused on follow-up outcomes
S2	2	Descriptive analysis
C2	2	Useful for context; limited clinical outcome significance
Total	19	—
Level	3	—

Literature Summary:

This survey-based study describes real-world clinical practice patterns regarding ureteral stent placement after routine ureteroscopic stone surgery. The findings show variability in stenting decisions, reflecting that stent use is influenced by procedural complexity, perceived risk of postoperative obstruction, ureteral trauma, and surgeon preference. The study supports the clinical background that ureteral stents are widely used as an adjunct to ureteroscopy and

	BIORAD MEDISYS PVT. LTD.	Document No.	CER/DJS/61
		Revision. No	01
	CLINICAL EVALUATION REPORT PRODUCT- DOUBLE J STENT	Revision Date	26-12-2025

stone treatment, but that routine use is not uniform across practitioners. For CER state-of-the-art discussion, this survey demonstrates that ureteral stenting is an established and commonly accepted intervention, while also supporting the concept of selective, indication-driven use. It reinforces that clinical management relies on physician judgement and that stent placement decisions incorporate balancing expected benefits (ensuring drainage) against known side effects (symptom burden and discomfort).

The dilemma of post-ureteroscopy stenting (Hughes B et al., 2014)

Parameter	Score	Justification
D2	2	Contextual discussion of stenting practice
A1	3	Relevant to intended clinical use scenario
P1	3	General ureteroscopy population
R2	2	Discusses benefits/limitations rather than providing new outcomes
T3	1	Commentary / narrative discussion
O2	2	Qualitative outcomes
F2	2	Not outcome-follow-up based
S3	1	No statistical analysis (not applicable)
C2	2	Useful contextual clinical reasoning
Total	18	—
Level	4	—

Literature Summary:

This publication discusses the clinical rationale and ongoing debate regarding ureteral stenting following ureteroscopy. It highlights that stents are frequently used to prevent obstruction caused by ureteral edema or trauma and to ensure drainage, but they can also introduce predictable patient discomfort and urinary symptoms. The paper emphasizes that clinical decision-making should consider the patient's risk profile and procedural findings rather than applying routine stenting universally. The source contributes to CER content by providing a clinically grounded interpretation of the evidence base and by contextualizing the accepted trade-offs associated with ureteral stent placement. It supports a state-of-the-art conclusion

	BIORAD MEDISYS PVT. LTD.	Document No.	CER/DJS/61
		Revision. No	01
	CLINICAL EVALUATION REPORT PRODUCT- DOUBLE J STENT	Revision Date	26-12-2025

that ureteral stents are effective and widely used, while acknowledging that symptom burden is an inherent limitation that must be managed through appropriate selection, patient counselling, and timely removal.

Safety and efficacy of ureteral access sheaths (Stern JM et al., 2007)

Parameter	Score	Justification
D2	2	Focus is on access sheath (adjunct), not stent itself
A2	2	Relevant procedural context where stents may be used
P1	3	Ureteroscopy population relevant
R2	2	Provides procedural safety data, indirect to stents
T2	2	Clinical study type (varies; generally observational)
O2	2	Outcomes relate to access sheath safety more than stent
F1	3	Appropriate peri-procedural assessment
S2	2	Appropriate analysis for reported endpoints
C2	2	Contextual relevance (procedure-related)
Total	20	—
Level	3	—

Literature Summary:

This study evaluates the safety and effectiveness of ureteral access sheaths used during ureteroscopy. While the device under study is not a ureteral stent, the publication provides relevant procedural context for ureteroscopic interventions, which frequently involve temporary ureteral stenting before or after the procedure. The paper addresses procedural factors such as ureteral trauma, access-related complications, and technical performance considerations that can influence postoperative management decisions, including whether a ureteral stent is placed to support drainage and healing. For CER purposes, the study contributes indirect evidence relating to the clinical environment and procedural risk context in which ureteral stents are used. This supports a holistic state-of-the-art assessment by linking stenting practices to ureteroscopic technique and associated risks.

	BIORAD MEDISYS PVT. LTD.	Document No.	CER/DJS/61
		Revision. No	01
	CLINICAL EVALUATION REPORT PRODUCT- DOUBLE J STENT	Revision Date	26-12-2025

The Post-Ureteroscopic Lesion Scale (PULS): inter-rater reliability (Schoenthaler M et al., 2014)

Parameter	Score	Justification
D2	2	Focus is on lesion scale (procedure-related), not stent
A2	2	Relevant to ureteroscopic injury assessment context
P1	3	Ureteroscopy population relevant
R2	2	Provides injury grading framework (indirect to stent risk context)
T2	2	Multicenter evaluation study
O2	2	Outcomes are reliability/scale-related
F1	3	Appropriate assessment timeframe
S1	3	Statistical reliability analysis performed
C2	2	Supports consistent reporting of ureteral lesions
Total	21	—
Level	3	—

Literature Summary:

This multicenter study evaluates the inter-rater reliability of the Post-Ureteroscopic Lesion Scale (PULS), a standardized grading system for ureteral lesions following ureteroscopy. Although the publication does not evaluate ureteral stents directly, it provides an important framework for consistent classification of ureteroscopic injury severity, which is relevant to clinical decision-making regarding postoperative measures such as ureteral stenting. The study supports the clinical background that ureteroscopic injury can vary in severity and that structured assessment improves consistency in reporting and management. From a CER standpoint, PULS is relevant as it helps contextualize stent use as part of post-procedural management strategies aimed at ensuring drainage and supporting ureteral healing when lesions are present. The publication therefore contributes indirectly to the state-of-the-art understanding of ureteroscopy-related risks and their relationship to stenting decisions.

	BIORAD MEDISYS PVT. LTD.	Document No.	CER/DJS/61
		Revision. No	01
	CLINICAL EVALUATION REPORT PRODUCT- DOUBLE J STENT	Revision Date	26-12-2025

4.5 Data analysis

The data held by the manufacturer for the subject device is used to demonstrate compliance with the MDR's general safety and performance requirements. The following data sources are considered for the analysis:

- 1. Pre-clinical data** (Biocompatibility, Bench testing) in section 4.3.1 are used as evidence for the safety and performance of the subject device.
- 2. Clinical data** (similar device) sections 4 and 4.2.1 are used as evidence for the acceptability of the subject device's safety, performance, and side effects.
- 3. State-of-the-art literature data** in sections 3 and 4.4 is used as supportive evidence for the appropriateness of the outcome parameter and to derive safety and performance threshold values. Literature data is also analysed to compare the benefit-risk of the subject device with the alternative treatment options.
- 4. Risk Management data** is used as evidence for the safety of the subject device.

4.5.1 Requirement on safety

The device's safety and performance data include preclinical and literature data of similar devices (state-of-the-art literature data), adverse event database monitoring data of similar devices, and risk management data.

Table 28: Safety and performance indicators identified from the literature

Category	Parameter	Relevance Based on Reviewed Literature
Performance	Urinary Drainage Restoration	Identified across systematic reviews (Haleblian; Nabi; Song) as the primary intended purpose of ureteral stents—maintenance of ureteral patency and prevention of obstruction after ureteroscopy or in stone disease. Clinical success defined by absence of hydronephrosis, preserved renal function, and lack of obstructive symptoms.

	BIORAD MEDISYS PVT. LTD.	Document No.	CER/DJS/61
		Revision. No	01
	CLINICAL EVALUATION REPORT PRODUCT- DOUBLE J STENT	Revision Date	26-12-2025

	Symptom Relief Score	Joshi et al. demonstrated measurable symptom burden using validated tools (USSQ). Literature confirms reduction in obstruction-related flank pain but highlights stent-related LUTS. Symptom scores are critical in benefit–risk evaluation.
	Reduction in Reintervention Rate	Meta-analyses (Song; Nabi) evaluated unplanned visits and need for secondary procedures. Appropriate stent use reduces postoperative obstruction-related reinterventions in indicated patients.
	Time to Obstruction Recurrence	Discussed in retained-stent and encrustation literature; prolonged indwelling increases risk of obstruction due to encrustation or migration. Short-term use supports lower recurrence risk.
	Technical Success Rate	Systematic reviews report high placement success (>95%) when performed by trained urologists. Technical feasibility well established in routine practice.
	Procedural Time	Reviewed indirectly in ureteroscopy-related studies (Song; Stern). Stent placement is a short adjunctive procedure and does not significantly prolong operative time when routinely performed.
	Migration Rate	Case reports and reviews document migration as an uncommon but recognized complication (<5% in literature). Proper sizing and positioning reduce occurrence.
	Encrustation Rate	Extensively described in forgotten/retained stent literature. Risk correlates strongly with dwell time. Short-term stenting significantly reduces encrustation risk.

	BIORAD MEDISYS PVT. LTD.	Document No.	CER/DJS/61
		Revision. No	01
	CLINICAL EVALUATION REPORT PRODUCT- DOUBLE J STENT	Revision Date	26-12-2025

	Patient Tolerance / Discomfort	Strongly supported by Joshi et al. and subsequent analyses. Irritative symptoms common but generally transient and manageable.
	Radiographic / Endoscopic Visibility	Highlighted in management of retained or complicated stents. Radiopacity supports safe placement verification and retrieval, improving procedural safety.
Safety	Hematuria (Minor/Major)	Frequently reported post-procedural effect; generally mild and self-limited (Haleblian; Nabi). Major bleeding is rare.
	Pain / Discomfort Score	Identified as common adverse effect (Joshi; Nabi). Typically transient and manageable; affects quality of life but not long-term safety.
	Urinary Tract Infection (UTI)	Reported complication in meta-analyses and retained-stent literature. Risk increases with prolonged indwelling duration; short-term use limits risk.
	Stent Migration	Documented in case reports and systematic reviews. Rare but may require repositioning or removal.
	Stent Encrustation	Strongly associated with prolonged dwell time (retained stent case reports). Early removal mitigates risk.
	Anesthesia-Related Events	Rare and procedure-dependent. Literature indicates most stents are placed during endourological procedures under standard anesthesia protocols.
	Need for Emergency Surgery	Rare; typically associated with neglected or complicated retained stents. Not characteristic of appropriate short-term use.
	Bladder Irritation / LUTS	Most common stent-related side effect (Joshi; Haleblian). Generally non-serious but clinically relevant to patient tolerability.
	Vesicoureteral Reflux (VUR)	Mechanistically possible due to bypass of anti-reflux mechanism; rarely

	BIORAD MEDISYS PVT. LTD.	Document No.	CER/DJS/61
		Revision. No	01
	CLINICAL EVALUATION REPORT PRODUCT- DOUBLE J STENT	Revision Date	26-12-2025

		clinically significant in short-term use per literature.
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4.5.2 Overall Safety and Performance analysis

The use of the Double J Stent (Short Term) has demonstrated a positive clinical impact in the management of ureteral obstruction and in providing temporary internal urinary drainage following urological procedures such as ureteroscopy, lithotripsy, and treatment of ureteral strictures or injuries. The primary therapeutic objective of the device is the restoration and maintenance of unobstructed urine flow from the renal pelvis to the bladder, thereby preventing hydronephrosis, relieving flank pain, and preserving renal function.

From a safety perspective, although appropriate precautions are implemented, several factors may influence patient outcomes during ureteral stent placement. These include the patient's overall medical condition, age, anatomical considerations, presence of infection, and the indication for stenting. Procedural factors such as correct stent size selection, accurate positioning within the ureter, and adherence to aseptic technique are critical to minimizing complications. The device is supplied sterile and intended for single use; therefore, proper handling and strict compliance with the Instructions for Use (IFU) are essential to reduce risks such as infection or device malfunction.

The clinical performance of the Double J Stent (Short Term) is dependent on appropriate patient selection and use according to the intended purpose. When properly placed, the stent maintains ureteral patency through its hollow lumen and pigtail retention design, reducing the risk of obstruction due to edema or residual stone fragments. Literature evidence confirms high technical success rates for placement and effective drainage in indicated patients.

Recognized side effects associated with ureteral stenting include transient hematuria, lower urinary tract symptoms (frequency, urgency, dysuria), flank discomfort, and occasional urinary tract infection. Migration and encrustation are reported in the literature but are uncommon

	BIORAD MEDISYS PVT. LTD.	Document No.	CER/DJS/61
		Revision. No	01
	CLINICAL EVALUATION REPORT PRODUCT- DOUBLE J STENT	Revision Date	26-12-2025

in short-term use and are strongly associated with prolonged indwelling duration. Severe complications are rare and typically related to delayed removal or improper use rather than inherent device design issues.

Based on the cumulative evaluation of scientific literature, equivalence data, risk management documentation, and post-market surveillance findings, the Double J Stent (Short Term) demonstrates acceptable safety and clinical performance consistent with the current state of the art. When used as intended and in accordance with the IFU, the device provides effective short-term urinary drainage with a favorable benefit–risk profile.

	BIORAD MEDISYS PVT. LTD.	Document No.	CER/DJS/61
		Revision. No	01
	CLINICAL EVALUATION REPORT PRODUCT- DOUBLE J STENT	Revision Date	26-12-2025

4.5.3 Requirement on acceptable benefit-risk profile

Table 29: Overall Rates of Performance and Safety parameters for similar device retrieved from literature

Performance Parameters	Range / Status	Safety Parameter	Range / Status
Technical success / Successful placement	High / routinely successful (reported as standard practice in ureteral stenting literature; generally >95% in routine clinical use)	Hematuria	Common; usually transient after placement / post-ureteroscopy (reported frequently in reviews; severity typically mild/self-limiting)
Urinary drainage restoration / Maintenance of patency	Effective in indicated cases (consistent conclusion across systematic reviews for obstruction/post-procedure drainage)	UTI	Uncommon to occasional in short-term post-procedure settings; higher with prolonged indwelling/retained stents
Symptom relief of obstruction-related pain	Clinically beneficial when obstruction is present, but can be offset by stent-related symptoms	Pain / irritative LUTS (frequency, urgency, dysuria)	Very common in indwelling stent patients (Joshi et al. + reviews consistently describe high symptom burden; commonly measured using USSQ/VAS)
Need for reintervention / unplanned intervention	Generally low in uncomplicated cases; indication-dependent (meta-analyses show routine stenting not always necessary post-uncomplicated ureteroscopy)	Migration / malposition	Uncommon; generally <5% and influenced by sizing/positioning; described as recognized complication
Time to obstruction recurrence	Strongly time-dependent; short-term use supports low recurrence, whereas prolonged retention increases risk	Encrustation / calcification	Time-dependent: low with short-term use, but can become significant with prolonged/forgotten stents (case reports/series show

	BIORAD MEDISYS PVT. LTD.	Document No.	CER/DJS/61
		Revision. No	01
	CLINICAL EVALUATION REPORT PRODUCT- DOUBLE J STENT	Revision Date	26-12-2025

			severe encrustation with months–years retention)
Radiographic / endoscopic visibility for follow-up & removal	Required/beneficial feature for positioning verification and retrieval (especially relevant in complicated/retained stent management literature)	Obstruction of stent (blockage)	Uncommon in short-term use; risk increases with encrustation, infection, or prolonged dwell time
Procedural feasibility (adjunct to ureteroscopy)	Feasible; standard adjunct step in endourology (survey + reviews support widespread routine capability)	Serious complications (e.g., perforation/erosion, sepsis, emergency surgery)	Rare; reported mainly in complicated cases, procedural injury contexts, or prolonged/retained stents rather than routine short-term use

Benefits of the device:

Table 30: Benefits and associated risks

Benefits of the Device (with references)	Associated Risks	Reported Rates / Evidence (from selected literature)
Restoration/maintenance of ureteral patency and urinary drainage in obstructive conditions and after endourological procedures; intended to prevent obstruction-related renal pain and facilitate drainage.	Irritative lower urinary tract symptoms (LUTS) (e.g., dysuria, frequency/urgency) due to indwelling stent irritation	After ureteroscopy, dysuria RR 2.25 (95% CI 1.14–4.43) and frequency/urgency RR 2.00 (95% CI 1.11–3.62) were higher in stented vs non-stented patients.
Reduction of postoperative ureteric obstruction and renal colic from ureteric edema/stone	Pain/discomfort and QoL impact (flank discomfort, bladder irritation affecting daily activities)	Indwelling stents are associated with a reduced quality of life in up to ~80% of patients (patient-reported impact).

	BIORAD MEDISYS PVT. LTD.	Document No.	CER/DJS/61
		Revision. No	01
	CLINICAL EVALUATION REPORT PRODUCT- DOUBLE J STENT	Revision Date	26-12-2025

manipulation; supports postoperative recovery where clinically indicated.		
Facilitation of passage of residual fragments via passive ureteric dilation with an indwelling stent (context of stone disease/ureteroscopy).	Hematuria (usually minor/self-limited) related to mucosal irritation/trauma	Reported as a recognized, generally transient post-stenting effect across ureteral stent literature; rates vary by procedure/setting (not consistently pooled across the selected set).
Potential reduction in unplanned medical visits/admissions in some post-ureteroscopy settings (although not statistically significant in pooled analysis).	Urinary tract infection (UTI) (risk influenced by patient factors, duration, peri-procedural context)	Meta-analysis found no significant difference in UTI rates between stented and non-stented groups after uncomplicated ureteroscopy (directionally comparable).
Established clinical utility in routine urology practice (well-characterized device category used widely for drainage/temporary internal bypass).	Migration / malposition (may impair drainage and require repositioning/removal)	Reported incidence in the literature typically ranges ~1–4.2% for proximal migration and ~2–10% overall migration depending on setting and technique.
Supports temporary internal bypass while definitive management proceeds, provided the device is removed within intended short-term duration.	Encrustation/calcification, fragmentation, obstruction (primarily time-dependent; increases with prolonged retention/forgotten stents)	Recognized complications; described as “not uncommon” in the context of stenting complications (risk increases with extended dwell time beyond intended use).

The likelihood of these risks occurring is low and consistent with accepted surgical practice. Importantly, no serious or life-threatening adverse events, long-term functional

	BIORAD MEDISYS PVT. LTD.	Document No.	CER/DJS/61
		Revision. No	01
	CLINICAL EVALUATION REPORT PRODUCT- DOUBLE J STENT	Revision Date	26-12-2025

complications, or device-related failures have been reported in the literature. No safety-related complaints or adverse trends have been observed in post-market surveillance or adverse event databases for the subject device.

Given the low incidence and minor severity of known risks, the Double J Stent (Short term) is considered clinically safe, with a favourable benefit-risk profile that supports continued use in the target population.

Table 34 and 35 provide a structured framework for evaluating the benefits and risks associated with a medical intervention. The benefits of the subject device focus on the positive aspects, considering factors such as the type, duration, probability, magnitude, availability, and patient perspective of the benefits. Risk evaluates the negative aspects, examining the severity, prevalence, duration, probability, relevance, and patient perspective of the risks.

The benefits and risks of the device are evaluated in the table below:

Table 31: Parameters for analysis of the benefits

Criteria	Description	Score (1–3)	Applicable Score
Type of Benefit	Restoration and maintenance of ureteral patency; prevention of obstruction; relief of hydronephrosis and flank pain; protection of renal function following ureteroscopic or obstructive conditions.	1 = Minor symptomatic benefit 2 = Moderate therapeutic benefit 3 = Significant therapeutic benefit (prevents obstruction-related complications)	3
Duration of Benefit	Benefit persists for the indwelling period (short-term use up to ~30 days), covering the critical postoperative or obstructive recovery phase.	1 = Very short (<1 week) 2 = Short-term (weeks) 3 = Long-term (>months)	2
Probability of Benefit	High probability of achieving intended urinary drainage when properly placed; literature	1 = Low (<50%) 2 = Moderate (50–80%) 3 = High (>80%)	3

	BIORAD MEDISYS PVT. LTD.	Document No.	CER/DJS/61
		Revision. No	01
	CLINICAL EVALUATION REPORT PRODUCT- DOUBLE J STENT	Revision Date	26-12-2025

	supports high technical success rates and predictable clinical function.		
Magnitude of Benefit	Prevents potentially serious consequences of obstruction (renal impairment, infection, pain); enables continuation of definitive treatment.	1 = Minimal 2 = Moderate 3 = Clinically significant / organ-preserving	3
Treatment Availability	Alternative treatments exist (e.g., percutaneous nephrostomy, selective non-stenting), but internal stenting remains standard and less invasive for indicated patients.	1 = Many superior alternatives 2 = Alternatives available but device remains standard 3 = No viable alternative	2
Patient Perspective	Although associated with transient urinary symptoms, patients benefit from obstruction relief and avoidance of external drainage devices. Quality-of-life impact is acceptable for short-term use.	1 = Low perceived benefit 2 = Acceptable benefit despite discomfort 3 = Highly favorable from patient perspective	2

Table 32: Parameters for analysis of the risks

Criteria	Description	Score (1–3)	Applicable Score
Severity of the Medical Condition	Ureteral obstruction may lead to hydronephrosis, infection, renal impairment, and severe pain if untreated.	1 = Mild condition 2 = Moderate condition 3 = Serious / potentially organ-threatening condition	3
Disease Prevalence	Ureteral obstruction and stone-related conditions are common in urological practice.	1 = Rare 2 = Occasional 3 = Common	3

	BIORAD MEDISYS PVT. LTD.	Document No.	CER/DJS/61
		Revision. No	01
	CLINICAL EVALUATION REPORT PRODUCT- DOUBLE J STENT	Revision Date	26-12-2025

Duration of Events (Risk Duration)	Most stent-related adverse events (e.g., LUTS, mild hematuria) are transient and resolve after removal; serious complications are uncommon and usually linked to prolonged retention.	1 = Very short/self-limiting 2 = Short-term and manageable 3 = Long-term or permanent	2
Probability of Event	Common events include transient LUTS and mild discomfort; serious complications (migration, encrustation, perforation) are uncommon and time-dependent.	1 = Rare (<5%) 2 = Occasional (5–20%) 3 = Frequent (>20%)	2
Relevance of the Risk to the Device	Most risks (pain, LUTS, hematuria) are inherent to the device category and procedure; severe risks are often related to prolonged indwelling or procedural factors rather than device defect.	1 = Indirect/low device relation 2 = Device-related but predictable/manageable 3 = Direct high-risk intrinsic to device	2
Patient Perspective	Patients may experience temporary discomfort; however, avoidance of obstruction-related complications and preservation of renal function generally outweigh short-term symptoms.	1 = Poor tolerability 2 = Acceptable tolerability 3 = High tolerability	2

The overall risk profile of the Double J Stent (Short Term) is considered **moderate and predictable**, with most adverse events being transient, manageable, and well documented in the literature. Severe complications are rare and primarily associated with improper use or prolonged indwelling duration beyond intended short-term use.

	BIORAD MEDISYS PVT. LTD.	Document No.	CER/DJS/61
		Revision. No	01
	CLINICAL EVALUATION REPORT PRODUCT- DOUBLE J STENT	Revision Date	26-12-2025

When balanced against the high probability and magnitude of therapeutic benefit (restoration of urinary drainage and prevention of renal complications), the overall benefit–risk profile remains favorable.

4.5.4 Requirement on acceptability of side-effects

Clinical data provides a comprehensive understanding of the application of the Double J Stent/Kit (Short Term) in biopsy sample collection. As part of PMS, adverse event databases have been searched, and no serious incidents related to the use of the Double J Stent/Kit (Short Term) in general have been reported. For literature analysis it can be concluded that the Double J Stent/Kit (Short Term) procedure is a generally safe and minimally invasive procedure, with moderate complication reported by most patients.

Though the device being biocompatible does not impose any risk of side-effects, issues like pain and discomfort can occur due to the procedure or anaesthesia.

Overall, the product has been well-received and demonstrates the effectiveness as claimed, with no reported allergic reactions, or serious adverse events in literature. Other than the risks discussed above in sub-section 4.6.5, no other side-effects have been reported for the Double J Stent/Kit (Short Term).

5 Conclusion

The clinical evaluation of the Double J Stent (Short Term) designed and manufactured by Biorad Medisys Private Limited demonstrates conformity with the applicable provisions of Regulation (EU) 2017/745 (EU MDR), including the relevant General Safety and Performance Requirements (GSPRs) set out in Annex I.

The evaluation was conducted in accordance with Article 61 and Annex XIV (Part A) of the MDR and was based on a systematic review of scientific literature, demonstration of equivalence with a marketed Double Pigtail Ureteral Stent, device-specific verification and validation data, risk management documentation, and post-market surveillance information. Based on the cumulative assessment of available clinical evidence, the following conclusions are drawn:

	BIORAD MEDISYS PVT. LTD.	Document No.	CER/DJS/61
		Revision. No	01
	CLINICAL EVALUATION REPORT PRODUCT- DOUBLE J STENT	Revision Date	26-12-2025

- The **Double J Stent (Short Term)** achieves its intended purpose of restoring and maintaining ureteral patency and facilitating temporary internal urinary drainage in indicated patients.
- The device demonstrates a **high technical success rate** and predictable clinical performance consistent with the current state of the art.
- The identified side effects—such as transient hematuria, irritative lower urinary tract symptoms, flank discomfort, urinary tract infection, migration, and encrustation—are well documented in the literature as inherent to the ureteral stent device category and are generally mild to moderate, manageable, and time-dependent.
- No new or unexpected safety concerns have been identified from the reviewed literature or post-market data.
- The device does not compromise the clinical condition or safety of patients, users, or other persons when used in accordance with the Instructions for Use and within the intended short-term duration.
- The overall benefit–risk profile is favorable, as the clinical benefits of relieving obstruction, preventing hydronephrosis, and preserving renal function outweigh the predictable and manageable risks associated with short-term ureteral stenting.

In conclusion, the available clinical evidence confirms that the Double J Stent (Short Term) is safe and performs as intended for its defined clinical indications. The device is considered to comply with the relevant GSPRs of EU MDR 2017/745 throughout its intended lifetime. Ongoing compliance is supported through the manufacturer’s Post-Market Surveillance and, where applicable, Post-Market Clinical Follow-Up activities, ensuring continued monitoring of safety and performance over the device lifecycle.

6 Date of the next evaluation

This clinical evaluation report will be updated every two years, with the next evaluation scheduled for December 2027. However, the evaluation may be conducted sooner if new information related to risks, design, or regulatory requirements emerges.

	BIORAD MEDISYS PVT. LTD.	Document No.	CER/DJS/61
		Revision. No	01
	CLINICAL EVALUATION REPORT PRODUCT- DOUBLE J STENT	Revision Date	26-12-2025

7 Qualification of responsible evaluators

Table 33: Qualification of responsible evaluators

NAME	QUALIFICATION & EXPERIENCE	ROLE
Dr. Prashant Pattnaik	Education: MBBS, M.S, (Surgery), M.Ch.(Urology) Bombay University Experience: - Professional Experience in reviewing and approving Clinical Evaluation based on MEDDEV 2.7/1 Revision 4. - Successfully managed the 1st Lithotripsy Centre of India soon after its inception until recent past considered as pioneer in the field in Asia. (Considered pioneer in Lithotripsy and advanced Endourology in the country) - Have unique experience of treating record number of (>50,000) of stone and allied diseases all over the country and overseas through Lithotripsy and Endourological procedures. (Holds lots of records and awards) - First to treat stone patients with pacemakers. Youngest, oldest and even excessively obese patients have been treated through ESWL and Endourology. - Treated several prostate diseases by Holmium Laser, TUNA, Plasma Kinetic Superpulse etc. (Introduced 1st in India) in addition to other conventional urogenital surgeries.	Independent review of clinical report

	BIORAD MEDISYS PVT. LTD.	Document No.	CER/DJS/61
		Revision. No	01
	CLINICAL EVALUATION REPORT PRODUCT- DOUBLE J STENT	Revision Date	26-12-2025

	Presented and published several articles at national and international levels.	
Sudhakar Saxena	Education: BSc & MBA Marketing. Approved in Microbiological & Chemical Testing from FDA. Technical Auditor for Cardiovascular Devices for CE Certification with NB “Institute for Testing and Certification” having NB No. 1023. Experience: He has been associated with Biorad from 8 years and he is Head of Regulatory department where he works with Clinical Evaluation Reports, Risk Management Reviews and Customer Complaints. He has attended training with regard to the use of Double J Stent/Kit (Long Term) . By virtue of qualification and work experience he has the ability to co-relate engineering aspects of the design and performance of the medical device with their clinical implications and hence has been an active contributor to the Clinical Evaluation Reviews and Report.	Approval of clinical report

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	BIORAD MEDISYS PVT. LTD.	Document No.	CER/DJS/61
		Revision. No	01
	CLINICAL EVALUATION REPORT PRODUCT- DOUBLE J STENT	Revision Date	26-12-2025

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10. Other references

Instructions for use (IFU) of Black Silicone Filiform Double Pigtail Ureteral Stent (Cook Medical).

Appendices

Appendix I: Literature Search Protocol

Appendix II: Literature Search Report

Appendix III: Declaration of Interest

Appendix IV: CVs of Evaluators

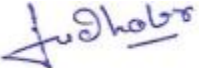
Appendix V: Adverse Events Search Results.

	BIORAD MEDISYS PVT. LTD.	Document No.	CER/DJS/61
		Revision. No	01
	CLINICAL EVALUATION REPORT PRODUCT- DOUBLE J STENT	Revision Date	26-12-2025

Revision history

Revision Number	Date	Description
00	26.12.2023	First Issue
01	26.12.2025	Periodic Revision

Document approvals

	Author	Independent Reviewer	Approver
Name	Awantee Kulkarni	Dr. Prashant Pattnaik	Sudhakar Saxena
Signature			
Date	26.12.2025	26.12.2025	26.12.2025
Title	Clinical Expert	Consultant - Medical Practitioner (External Evaluator)	General Manager Regulatory & Compliance